China Daheng Group, Inc. Beijing Image Vision Technology Branch

# User Manual

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DAHENG IMAGING 大恒图像

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# Preface

We really appreciate your choosing of DAHENG IMAGING products. GalaxyView is an application that used to debug and develop industrial camera. It applies to GigE, USB3.0, USB2.0, CXP interface industrial cameras, which supports preview in real time, parameter configuration, image acquisition and etc.

This manual introduce the installation and usage of GalaxyView, please be careful to read this manual before using.

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# 1. Introduction

## 1.1. Product Introduction

GalaxyView is a software application developed for debugging industrial cameras. It is applicable to GigE, USB3.0, USB2.0, CXP interface area cameras, and supports real-time preview, parameter configuration, image acquisition and other functions.

## 1.2. Symbolic Convention

See document symbols instruction as follows:

| Symbols | Description |
|---|---|
|  | **Instruction:** Supplement and interpretation of the text. |
|  | **Note:** Potential risks, may cause injury accidents, equipment damage or business interruption. |
| | **Warning:** Potential risks, may cause injury accidents, equipment damage or business interruption. |
| | **Danger:** High potential risks, may cause serious danger of casualties. |

## 1.3. Main Functions

- Easy installation, with no other drivers to operate
- Support multi-platform, compatible with Windows7/10/11 32/64 bits OS
- User friendly interface design with better interaction experience, simple operation steps,intuitive functions to maximum simplify the process
- Highlights the three core functions of Camera Control, Screen Preview and Attribute Configuration
- Support muti-cameras to shoot or preview images and enable fast screen switching
- Highly integrated tools to realize setting and reading of cameras and PC

## 1.4. Operating Environment

GalaxyView Installation and PC minimum configuration requirements are as follows:

| | |
|---|---|
| Operating System | Windows7/10/11 (32/64 bits, CN/EN OS) |
| CPU | Intel Core 2 Duo, 2.4GHz or higher |
| Memory | 2GB or higher |
| GPU | Support 1024 × 768 or higher resolution |

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| | |
|---|---|
| NIC | Recommendation: Intel Pro1000, I210 and I350 series |
| Cable | CAT-5e or CAT-6, length less than 100 meters |
| USB3 | Support USB3.0 interface |

Table 1-1 Operating environment

**Instruction:**

1. GalaxyView has already integrated hardware needed drivers, no need for others.
2. Resolution of your PC is 1024×768 or above, otherwise the window will not be displayed completely.
3. In order to avoid unknown anti-virus software block your way, better to add this application in the white list or close your anti-virus software.
4. This environment is the minimum configuration for the software's normal running. If users require high-definition, high-speed or multi-camera functions, it is necessary to improve the PC configuration according to the scene to ensure the running effect.

## 1.5. Main Page

After opening the GalaxyView, you will see:

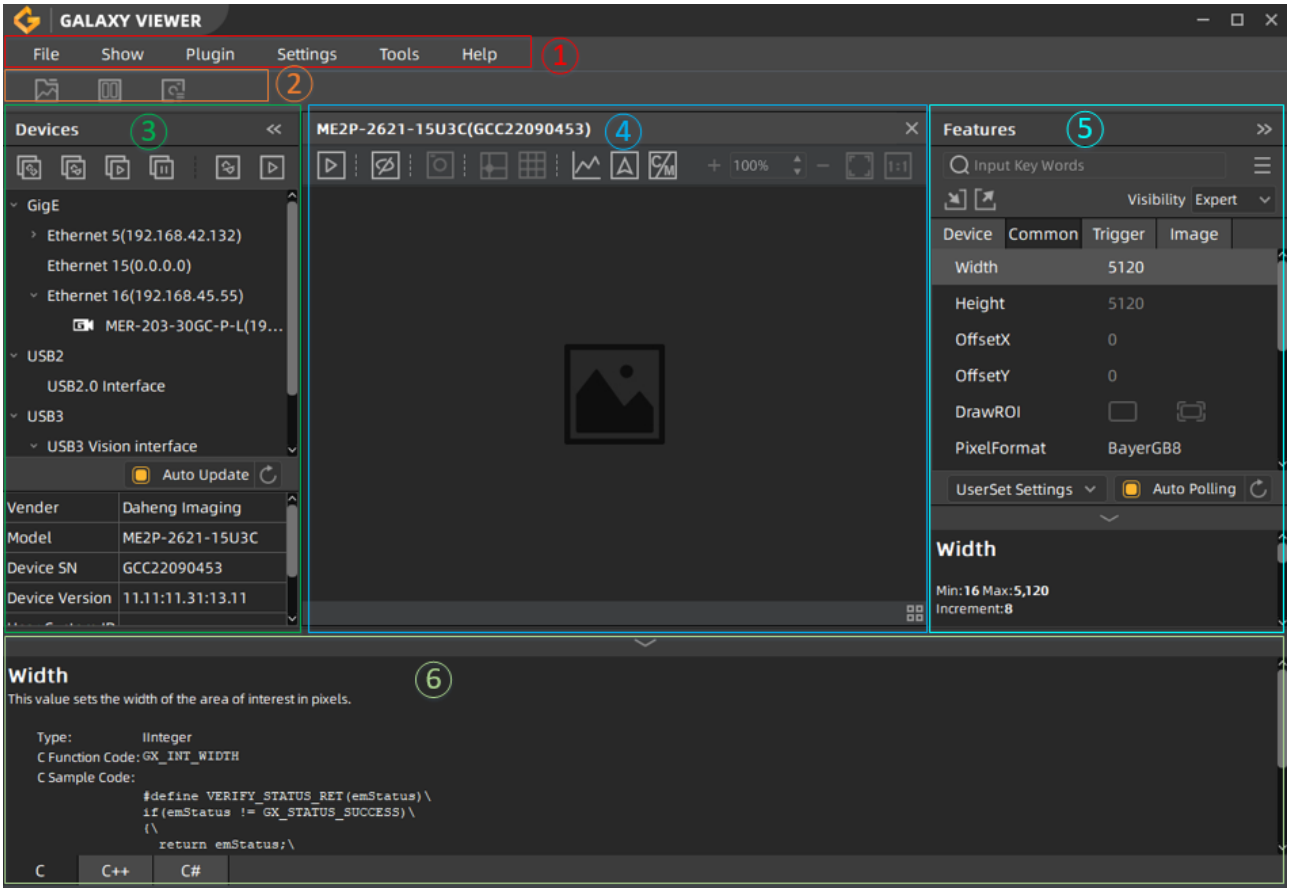



Figure 1-1 Main page of the application

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**Instruction:** GalaxyView's tab key traversal function between windows is not very complete yet.

Functional modules description:

| No. | Function | No. | Function |
|---|---|---|---|
| 1 | Menu bar | 4 | Preview Window |
| 2 | Toolbar | 5 | Device Features |
| 3 | Devices | 6 | Feature Document |

Table 1-2 Functional modules

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# 2. Environment Configuration

## 2.1. GigE Cameras Configuration

Before using the GigE camera, please make sure that the PC and the camera are in the same LAN, and set the PC NIC MTU (Maximum Transmission Unit) as Jumbo Frame mode.

Step1: Make sure that the camera power and transmission cable are in in good conditions.

Step2: Click Windows Start Menu to find NICConfigTool.exe and ensure that the Jumbo Frame is selected.

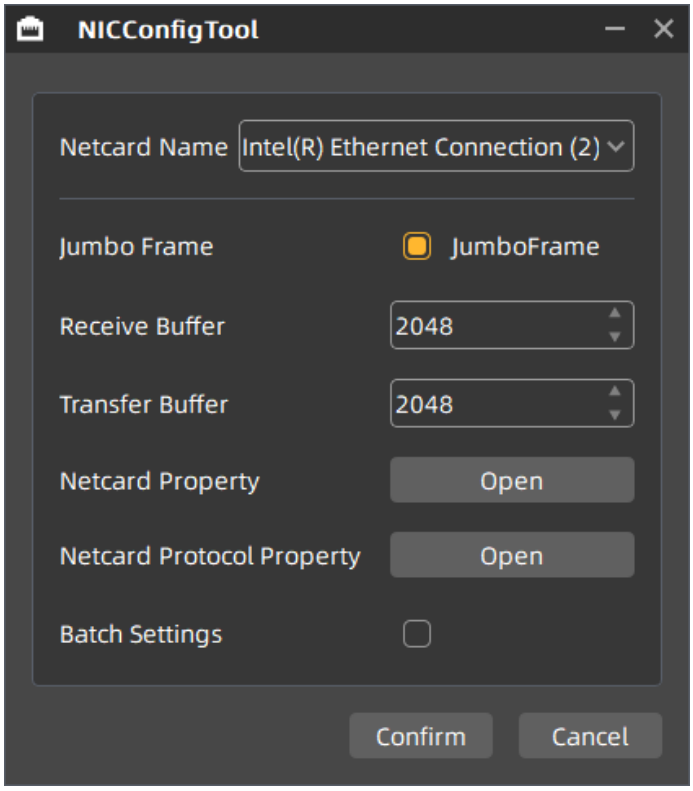



Figure 2-1 NICConfigTool

**Instruction:**

1. The package while installing will select all NIC Jumbo Frame by default.
2. If failed in setting the Jumbo Frame through NICConfigTool.exe, then click Netcard Property button to set the Jumbo Frame as 9014 bytes or 9KB, if failed again or Jumbo Frame cannot be found, please update your NIC or replace.

Step3: Set the IP address of the PC network port. Click Netcard Protocol Property to set IP address. Static IP is recommended.

Step4: Click Devices to search the GigE cameras. If the camera cannot be connected, please double click the camera and the GxGigEIPConfig (IP configuration tool) will show up, on which users can set the camera IP address (Static IP recommended).

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## 2.2. USB Cameras Configuration

Please be sure about that your PC is already configured with USB drive before using, or you will not find the cameras if fails in drive installation.

When connecting with the cameras through the PC USB interface, windows will detect the hardware and install the USB driver automatically. You can check the USB driver is installed or not through GxDriverManager in Start menu, if the driver is not be Installed, please execute the GxDriverManager tool.

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# 3. Menu Bar

Set **File**, **Show**, **Settings**, **Tools** and **Help** in Menu Bar.

## 3.1. File

Click file to find **Open Image**, **Save Image**, **Import Device Settings**, **Export Device settings** and **Save Device XML** options.

Descriptions of every single functions are as follows:

- **Open Image**

Choose a local image to open in the Preview Window. Supporting image formats: JPG, BMP, PNG, TIFF, RAW. See details in 4.1.Preview.

- **Import Device Settings**

Choose one connected camera, Click “Import Device Settings”, and then pop up file selection window, just choose one configuration file to import, if fails, there will be a pop-up window to prompt.

 **Note:** Cameras in acquisition status cannot import device configuration.

- **Export Device Settings**

Choose one connected camera, Click “Export Device Settings” , and then pop up the saving window, just name the file and set the saving the path as local file, if fails, there will be a pop-up window to prompt.

 **Note:** Cameras in acquisition status cannot export device configuration.

- **Save Device XML**

Choose one connected camera, Click “Save Device XML”, and then pop up the saving window, just name the file and set the saving the path as local file, if fails, there will be a pop-up window to prompt.

## 3.2. Show

- **FPS**

The displayed frame rate when in acquisition. Users can choose 30fps (default) or 60fps.

- **Show Incomplete Frame**

Users can decide whether to show this data or not (not display by default).

 **Instruction:**

1. Show Incomplete Frame will display bad image in the window, you can hide the bad frame images by disable this function.
2. Show frame rate and incomplete frame are globally effective for all cameras.

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## 3.3. Plugin

Plugin is an extensible tool which contains a series of tools. These plugins can be loaded and unloaded through Plugin Manager. The plugins included are as follows:

| Plugin | Description |
|---|---|
| Image Processing Plugin | Include color correction, pixel correction and set contrast, saturation and sharpen value |
| Videos and Images Save Plugin | Save image stream and video stream continuously in real time |
| Lut Creat Tool Plugin | Create Look Up Table and import/export to the camera, save on local files |
| Flat Field Correction Plugin | Flat field correction |
| Static Defect Correction Plugin | Static defect correction |

Table 3-1 Plugin description

### 3.3.1. Plugin Manager

Plugin Manager is used to adjust the priority of the plugins and show plugin information

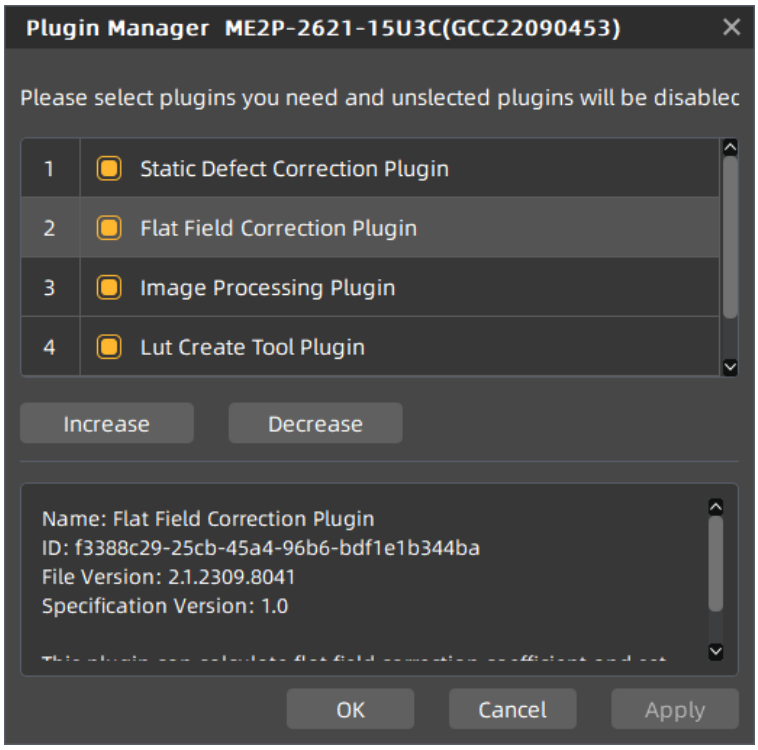



Figure 3-1 Plugin manager

- **Disable the plugin**

Select the plugins you need, and the unselected plugins are disabled.

- **Plugin details**

The plugin details will show below when being selected.

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- **Adjust plugin priority**

You can adjust the plugin priority by clicking Increase or Decrease.

### 3.3.2. Image Processing Plugin

Use Image Processing Plugin to correct image color and defect pixels, set contrast, saturation and sharpen values as follow:

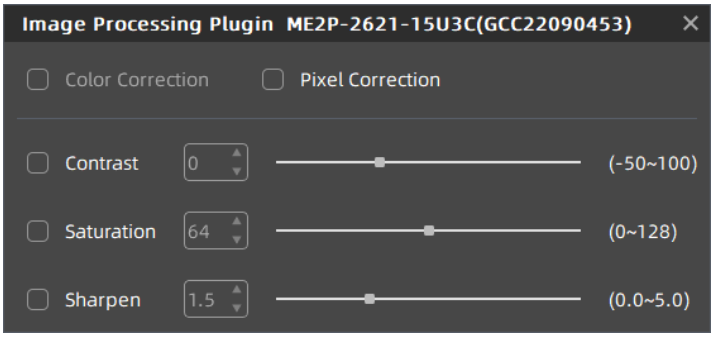



Figure 3-2 Image processing plugin

### 3.3.3. Videos and Images Save Plugin

Save image stream and video stream continuously in real time, see as follow:

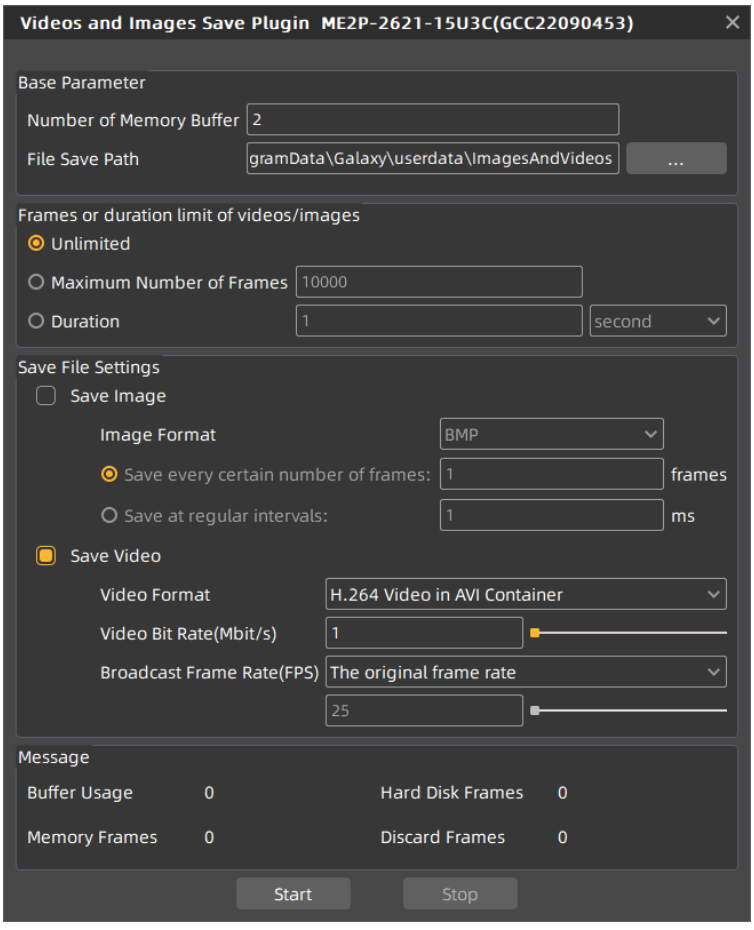



Figure 3-3 Videos and images save plugin

After setting every part, click Start to save images or videos, see details as follow:

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- **Base Parameter**

You can set the number of memory buffer and file save path here, the default path is: C:\ProgramData\Galaxy\userdata\ImagesAndVideos.

- **Frames or duration limit of videos/images**

You can choose unlimited mode or set a maximum number of frames or set a duration value.

- **Save File Settings**

There are two options with save image or save video:

1. Save Image
   a) Set image format (bmp, jpg, png, tiff, raw)
   b) Choose to save the image every frames or every millisecond
2. Save Video
   a) Set video format (avi, mp4)
   b) Set video bit rate (Mbit/s): 1~50
   c) Set broadcast frame rate (FPS) as original frame rate or custom frame rate

- **Message**

Show the details of your saved images or videos, which includes buffer usage, hard disk frames, memory frames and discard frames.

### 3.3.4. Lut Creat Tool Plugin

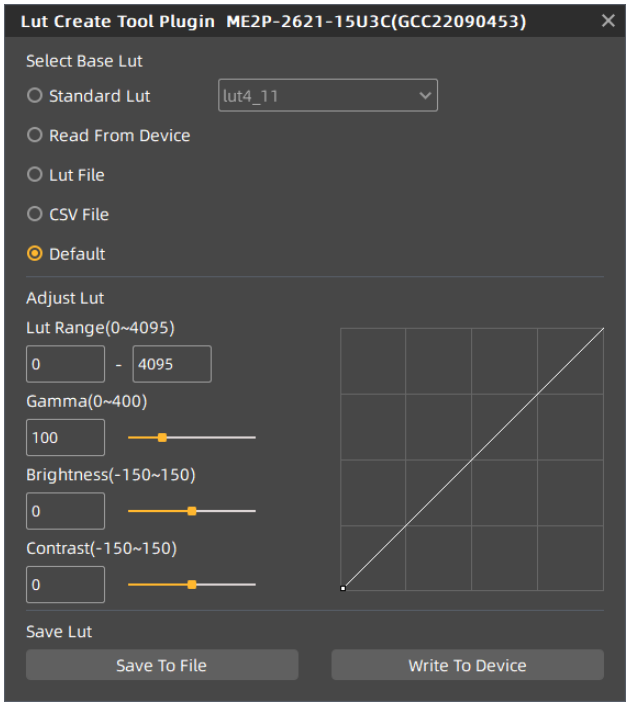



Figure 3-4 Lut create tool

The functions as follows:

1. Set Gamma, Brightness and Contrast values.
2. Read the saved Lut form device.
3. Write the adjusted Lut into the device.
4. Read the saved Lut form Lut/CSV file.
5. Save the adjusted Lut to file.

The initial default value and layout of Lut Create Tool Plugin are as Figure 3-4, see function description as follows:

**[Select Base Lut]** Include Standard Lut, Read From Device, Lut File, CSV File and Default options. There are eight Lut value in Standard Lut. Users can read the written Lut from the device, also, Lut/CSV file can read the saved value as well. It will be Default mode if no operating.

**[Adjust Lut]** Adjust the Lut range, Gamma, brightness and contrast to see the overlay effect base on the standard Lut.

**[Save Lut]** Write the currently generate Lut into device, save Lut/CSV file.

**[Line Chart]** Show the currently generate Lut in line chart.

- **Scenarious:**

After setting all parameters until you expected, click Write to Device button in order to recall the Lut you set next time. And the saved Lut will be write into Device UserSet0. All parameters you have set last time can be used by selecting Read From Device.

Another way to use the already set Lut is to save the Lut as file, select Lut File option to choose a Lut file you saved before, and all parameters will be recovered again in any device (For devices that don't support read/wirte Lut function or Lut can't be used situations).

- **Select Base Lut**

1. Standard Lut

There are eight Lut values in the drop down box, which are generated from factory preset in order to achieve the best image effect. When you choose different Standard Lut value, the Line Chart will change as well. You can adjust Lut Range, Gamma, Brightness and Contrast value to meet your expectation.

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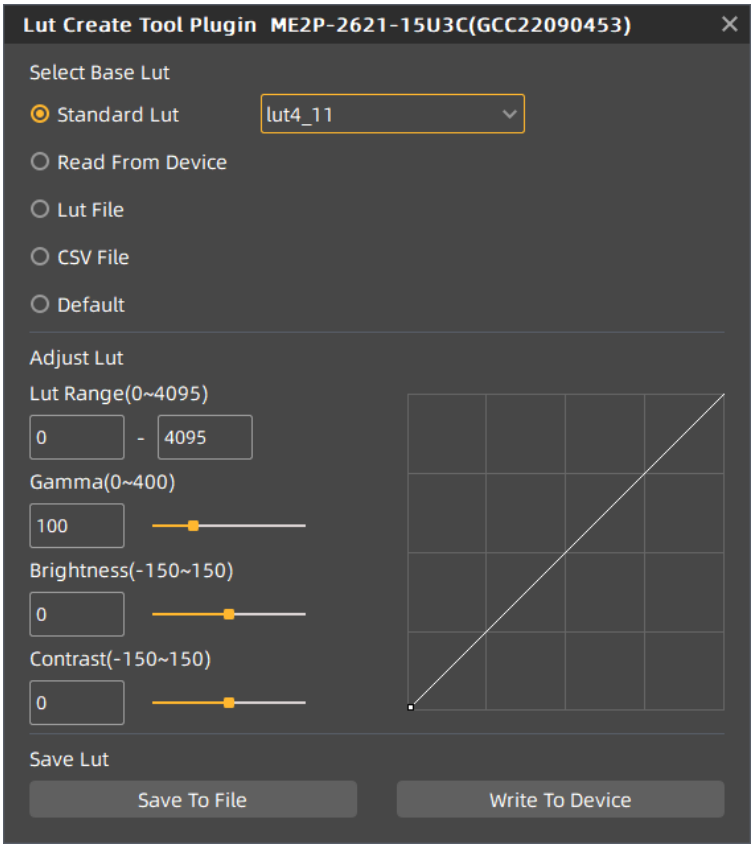



Figure 3-5 Standard Lut options

2. Read From Device

After selecting Read From Device, UserSet0 will be loaded automatically, and then, load the device saved Lut value. If the device support LUTEnable function, the value will be set as True automatically and users can see the image effect in real time.

When Read From Device is selected, the line chart and image effect are update to correspond with the new import Lut. For example, if you set Standard Lut as knee2, Lut Range as 0-1023, Gamma as 110, Brightness as 100, Contrast as 100, then click Write to Device, the next time you open Lut Create Tool Plugin window, you don't have to set all parameters above again, just select Read From Device and all set values will be restored.

3. Lut File

If you have saved an adjusted Lut file before, then select **Lut File** option to use the set data without resetting. Choose a .lut file on the pop up window.

4. CSV File

If you select CSV File, the line chart and image effect will be updated refer to the .csv file you chose. But it should be noted that when you select **CSV File**, all parameters in **Auto Create Lut** will be banned to adjust.

Users can modify .csv file manually. The csv storage format is: save the decimal number of every four bytes to the first cell of each line, the maximum number in each cell is 4095, and 4096 rows in total. The

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line chart is update according to the first row of every 16 rows, please abbey this rule to modify the data, or it will be failed in reading the file.

5. Default

It is **Default** (factory Lut data) mode by default if you don't choose any of the five options in **GroupBox**. Moreover, if any error occurs, it will switch to **Default** automatically, and the line chart will be diagonal.

- **Save Lut**

There are two options for users to choose: **Save To File** and **Write To Device**.

1. Save To File**:** Save the current Lut into local file (Lut/CSV)

   The default save path is: \resource\gxplugins\LookUpTable\Lut12.

2. Write To Device**:** Write the current Lut into UserSet0 and modify UserSetDefault as UserSet0, then the next time you read from device the UserSet0 will be loaded again.

- **Read Lut**

There are two methods to save the Lut file and configure into your cameras:

1. Lut Create Tool Plugin: Select **Lut File** to choose an exist .lut file, and then click **Write To Device**.

2. API: Users can get Lut file data by accessing read Lut interface in GxIAPI (Common programming interface library) and DxImageProc (Image processing algorithm interface library), parse the Lut data as camera configurable data, see details as follows:

   a) Get the camera Look up Table length.

   b) Apply Lut buffer resource according to the Lut length.

   c) Read the .lut file and get the Lut buffer data.

   d) Configure the Lut buffer data into the camera (Lut enabled).

   e) Write the current Lut into UserSet0 and modify UserSetDefault as UserSet0, then the next time you read from device the UserSet0 will be loaded again.

API supports C/C++/C#, see interface descriptions and sample code in "C SDK Programming Reference Manual" relevant section, "C++ SDK Programming Reference Manual" relevant section, "DotNET SDK Programming Reference Manual" relevant section.

### 3.3.5. Static Defect Correction Plugin

Static Defect Correction Plugin support all series of DAHENG IMAGING digital camera. The plugin is integrated into GalaxyView.exe. After opening the device through GalaxyView, open the Static Defect Correction plugin from the menu bar plugin list. Using the plugin, you can achieve the following functions:

1) Analyze the defect pixel in the current images of the device, including Bright dark scene and Actual scence.

2) Execute Static Defect Correction on the images.

3) Save the defect pixel information to the device.(The camera which support Static Defect Correction)

4) Save the defect pixel information to the file

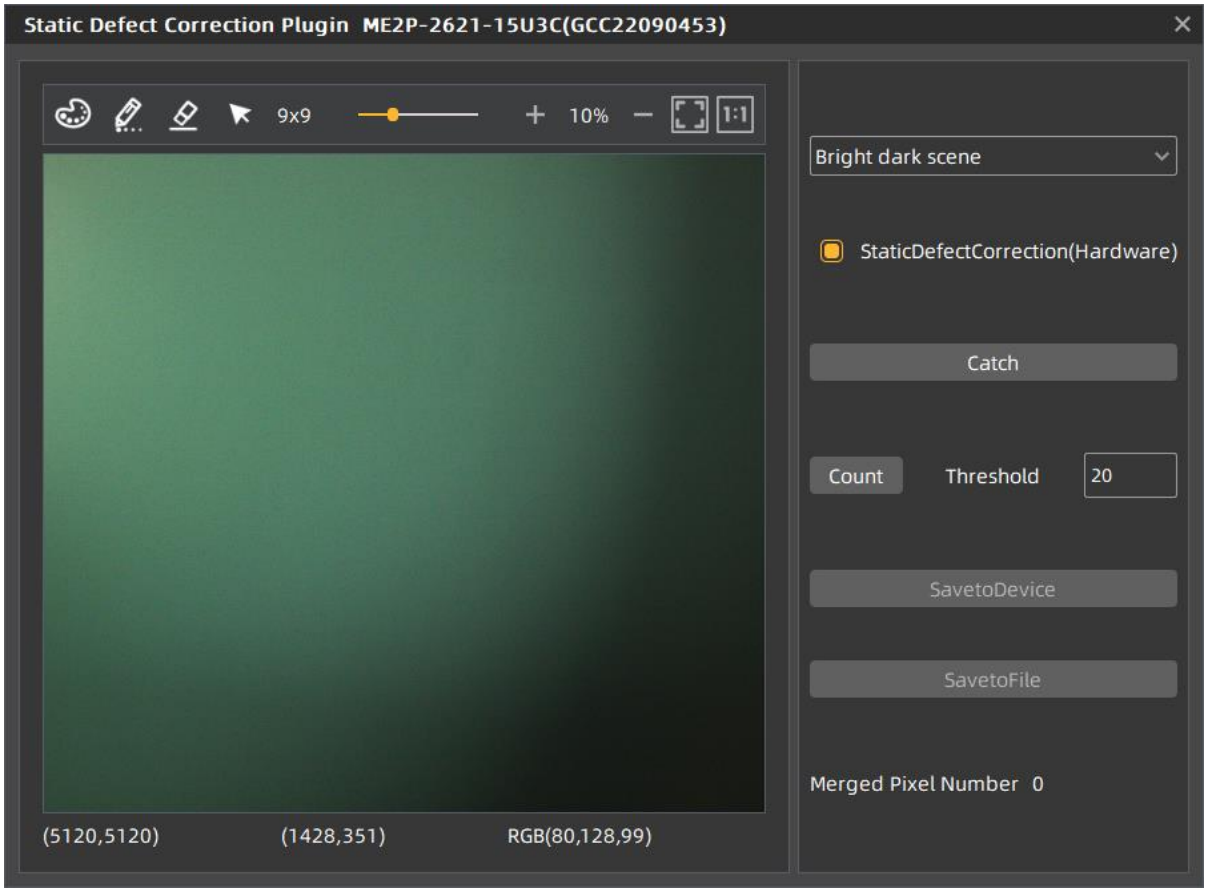



Figure 3-6 Static Defect Correction Plugin

After opening the device through GalaxyView.exe and opening the Static Defect Correction plugin, the initial state of the GUI is shown in Figure 3-6. The plugin layout and function description are as follows:

| No. | Widget | Function |
|---|---|---|
| 1 | Catch | Acquisition an image to analyze the location of the defect pixels and noise points |
| 2 | Threshold | Set the threshold for defect pixels and noise points judgment |
| 3 | Bright dark scene | Count the defect pixels |
| 4 | Actual scene | Count the noise points |
| 5 | Count | Count the location of the defect pixels and noise points |

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| | | |
|---|---|---|
| 6 | StaticDefectCorrection (hardware) | Choose whether to perform Static Defect Correction |
| 7 | SavetoDevice | Save the defects information to the device |
| 8 | SavetoFile | Save the defects information to a file |
| 9 | Image display area | Display the image. After counting the defect pixels and noise points, the location of the defect/noise pixels will be marked on the displayed image |
| 10 | Merged pixel number | Display the number of defects |
| 11 |  | Change the color of merged pixels |
| 12 |  | Manually mark the defects on the image |
| 13 |  | Erase the original merged pixels on the image |
| 14 |  | Set mouse gestures as arrow |
| 15 |   | Change the size of merged pixels |
| 16 |  | Zoom in |
| 17 |  | Zoom out |
| 18 |  | Adaptive present image |
| 19 |   | Present image 100% |

Table 3-2 Function description of the Static Defect Correction plugin

**[Image]** Capture an image through the catch button and display it in the white area in the middle of the plugin. The captured image is used to analyze the location of the defect pixels /noise points.

**[Defect pixel analysis]** User determines the range of defect pixels to be processed by setting the threshold and select the type of defect pixels. After clicking the "Count" button, the plugin will analyze the location of the defect pixels /noise points in the current image, and mark the location of the defect pixels on the image as red. The merged pixel number are display in the status bar.

If the current device supports the function of static defect pixel correction, and the merged pixel number is less than the number 8192 that the device supports. The defect pixel information will be written into the FLASH of the device.

**[Operation]** The Static Defect Correction can be performed by hardware or software. If the current device supports Static Defect Correction and the merged pixel number is less than 8192, hardware is preferred to perform Static Defect Correction, otherwise it can be executed through software. After the user check the Static Defect Correction box, the image displayed on GalaxyView is the image after performing the Static Defect Correction.

"SavetoDevice" button can save the defect pixel information to the FLASH of the device.

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"SavetoFile" button can save the defect pixel information to .dp or .csv file.

**[Image display area]** Display the image and the location of the defect pixels /noise points.

**[Status bar]** Show the defect pixels number/noise points number/merged pixel number.

3.3.5.1. Static Defect Correction Steps

1) Click the "Catch" to capture an image. For details, please see section “Acquisition Images”

2) Set threshold to determine the range of defect pixels

3) Check "Bright dark scene" or "Actual scene" to select the type of defect pixels

4) Click "Count" to complete the defect pixel analysis. The location of the defect pixels on the image will be marked and displayed in the status bar

5) Check "StaticDefectCorrection" to execute Static Defect Correction

6) When the device supports Static Defect Correction and the merged number is less than 8192, the user can through the "Save to Device" to write the statistics of the defect pixel information into the device, and it will still be valid after power off and restart

7) The user can click the "Save to File" to save the statistics of defect pixel information to a file. For details, please see section “How to Use Defect Pixel Data File”

3.3.5.2. Acquisition Images

1) When the device is in the stop acquisition mode, click "Catch", the image will be displayed in the GalaxyView acquisition GUI.

2) When the device is in the acquisition mode, click "Catch" to complete the image acquisition.

3) When counting defect pixels, it is required to acquisition images with uniform gray scale. For example, use bright field images when detecting dark defect pixels, and use dark field images when detecting bright defect pixels.

4) When the threshold is fixed, the number of defect pixels will be affected by the exposure time and gain. The greater the value of exposure time and gain, the greater the number of defect pixels.

5) The counted defect pixels by the device under the maximum resolution are applicable to any ROI image. The counted defect pixels by the device in the ROI are only applicable to the image in the ROI.

6) Click "  " to select the manual mark color.

7) Click "  ", set the mouse gestures as pencil to mark the defects on the image.

8) Click "  " to erase the original merged pixels on the image.

9) Click "  " to zoom in the image.

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10) Click "  " to zoom out the image.

11) Click "  " to adaptive present image.

12) Click "  " to present image 100%.

13) When performing image scaling operation, the current image scaling ratio will be displayed.

14) The current image width and height, pixel coordinates of mouse position and RGB value of mouse position are displayed in the status bar.

3.3.5.3. Static Defect Correction

1) "StaticDefectCorrection" is divided into "StaticDefectCorrection (software)" and "StaticDefectCorrection (hardware)"

2) When the device perform Static Defect Correction, the plugin will give priority to hardware to implement Static Defect Correction, which will be displayed as "StaticDefectCorrection (hardware)", otherwise it will be displayed as "StaticDefectCorrection (software)"

3) The condition for the device to perform Static Defect Correction is that the device supports the Static Defect Correction function and the number of defects is less than 8192.

4) When the device is in the acquisition mode, the user can check or uncheck "Static Defect Correction" to check the correction result.

**Instruction:** When the device performs Static Dead Pixel Correction, it is temporarily unable to remove the dead pixels at the left and right boundaries. The monochrome camera is 3 pixels from the boundary, and the color camera is 6 pixels from the boundary.

3.3.5.4. How to Use Defect Pixel Data File

1) The format of the defect pixel data file is ".dp" and ".csv", and the default save path is under the installation package directory:
*\Daheng Imaging\GalaxySDK\Demo\Win64\resource\gxplugins\DefectPixelCorrection;

2) When you need to use the SDK to implement the Static Defect Correction function, you can read the saved defect pixel data file and call the function of the image processing library: DxStaticDefectPixelCorrection to realize the Static Defect Correction of the image.

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### 3.3.6. Flat Field Correction Plugin

#### 3.3.6.1. Scenario Description

Depending on the generation and application locations of flat field correction coefficients, users may encounter the following usage scenarios:

| Scenario Type | Coefficient Generation | Coefficient Application | Camera Requirements |
|---|---|---|---|
| 1 | PC-side | PC-side | Compatible with all cameras |
| 2 | PC-side | Camera-side | Compatible with select cameras |
| 3 | Camera-side | Camera-side | Compatible with select cameras |

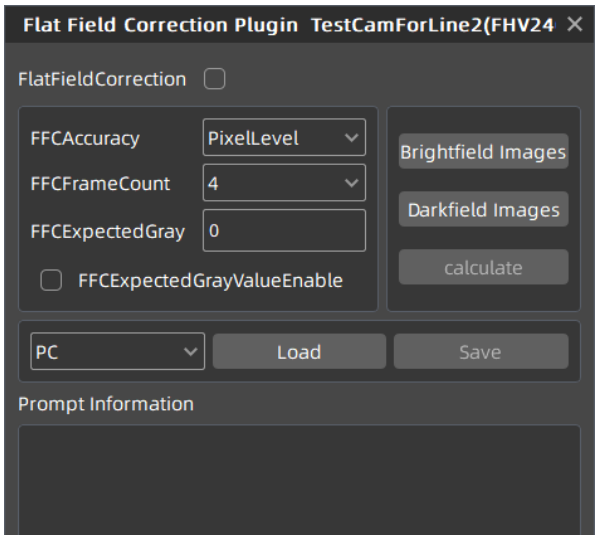



Figure 3-7 Scenario Type1

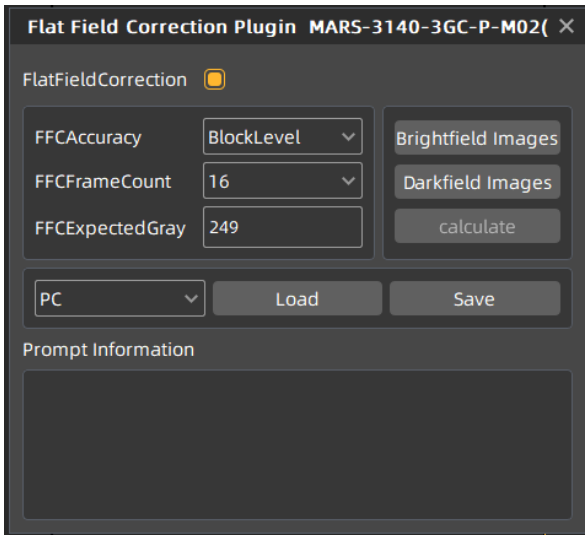



Figure 3-8 Scenario Type2 (MARS-3140-3GC-P-M02)

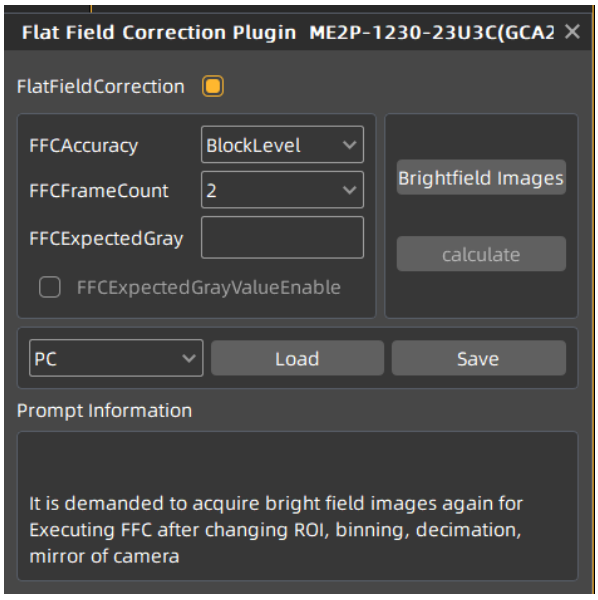



Figure 3-9 Scenario Type2 (Except MARS-3140-3GC-P)

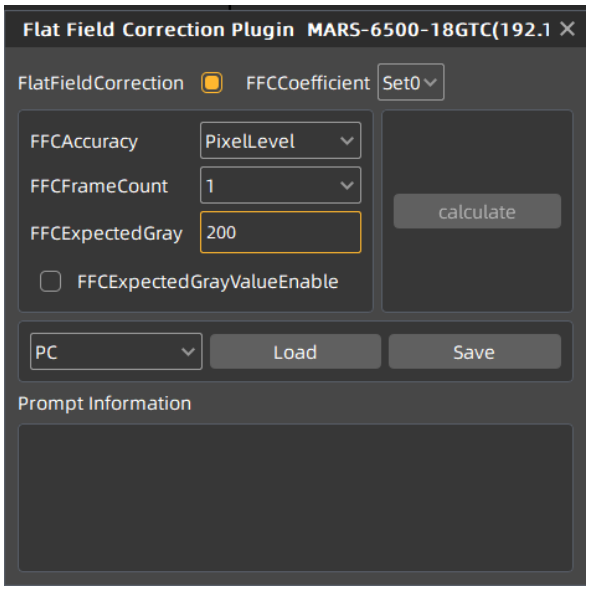



Figure 3-10 Scenario Type3

Functions:

1. Flat field correction.
2. Get the effective flat field correction coefficient from the device.
3. Write the effective flat field correction coefficient into device, in case of data loss caused by power off.
4. Obtain the saved flat field correction coefficient from file.
5. Save the effective flat field correction coefficient to file.

3.3.6.2. Camera Type Identification Method

Users can differentiate camera types based on the value obtained from querying ShadingCorrectionMode, as detailed below:

By querying the feature node through code:

C code example is provided below, for other languages, refer to the GxAPI Software Development Manual:

```
GX_STATUS emStatus = GX_STATUS_SUCCESS;
GX_NODE_ACCESS_MODE emAccessMode;
/*
When the value of emAccessMode is GX_NODE_ACCESS_MODE_NI or GX_NODE_ACCESS_MODE_NA
or GX_NODE_ACCESS_MODE_UNDEF, it indicates that the camera hardware does not support
the flat field correction function, and this condition corresponds to the scenario
type1.
*/

emStatus = GXGetNodeAccessMode(hDevice, "ShadingCorrectionMode", &emAccessMode);
```

```
GX_STATUS emStatus = GX_STATUS_SUCCESS;
GX_ENUM_VALUE stEnumValue;
emStatus = GXGetEnumValue(hDevice, "ShadingCorrectionMode", &stEnumValue);
/*
* 1. When the value of strShadingCorrectionMode is "FlatFieldCorrection" and this
condition corresponds to the scenario type2 (MARS-3140-3GC-P-M02 camera)
* 2 . When the value of strShadingCorrectionMode is "TailorFlatFieldCorrection"
and this condition corresponds to the scenario type2 (non-3140 model cameras).
* 3 . When the value of strShadingCorrectionMode is "DeviceFlatFieldCorrection"
and this condition corresponds to the scenario type3
*/string strShadingCorrectionMode = stEnumValue.stCurValue.strCurSymbolic;
```

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3.3.6.3. Layout and Function Description

The layout and function description of controls are as follows:

| UI Function | Scenario Type1 | Scenario Type2 | Scenario Type3 |
|---|---|---|---|
| FlatFieldCorrection | Controls whether to apply FFC to current image. When enabled, the GalaxyView application applies FFC coefficients. | Controls the camera's internal FFC function. | |
| FFCCoefficient | Not supported | | Supports multiple sets |
| FFCAccuracy | PixelLevel | BlockLevel | BlockLevel, PixelLevel |
| FFCFrameCount | Recommended value: 4. (Options: 1, 2, 4, 8, 16) — indicates the number of images merged during FFC calculation.<br>Larger values are advised for images with significant noise | | |
| FFCExpectedGray | User-defined expected gray value after FFC | | |
| FFCExpectedGrayValueEnabled | True: FFCExpectedGray value enabled<br>False: The algorithm automatically calculates the optimal gray value. | | |
| Brightfield Images | Compulsory | Compulsory | Not required |
| Darkfield Images | Optional | Optional | Not required |
| Calculate | Call the DxFFCCalculat interface to generate FFC coefficients for use in plugins. | Call the DxFFCCalculat interface to generate FFC coefficients, and load them into the camera's runtime memory. | Send the command "FFCGenerate" to the camera to trigger the calculation of FFC coefficients |
| PC: Load | Load a local FFC coefficient file (*.ffc) into the FFC plugin. | Load a local FFC coefficient file (*.ffc) into the camera's runtime memory | |
| PC: Save | Save the FFC coefficients calculated by the FFC plugin to a local *.ffc file. | Export the FFC coefficients from the camera's runtime memory to a local *.ffc file. | |
| Device: Load | Not supported | Load the FFC coefficients from the camera's non-volatile storage chip into the camera's runtime memory. | |
| Device: Save | Not supported | Save the FFC coefficients from the camera's runtime memory to the camera's non-volatile storage chip. | |

Table 3-3 Description of FFC function

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3.3.6.4. Execution Steps

- **General Steps**

1. Set the **FFCFrameCount** (Not necessary, you can skip to 2).
2. Click **Calculate** to execute the correction function. Upon successful completion, the **FlatFieldCorrection** will automatically enable.
3. Use the **FlatFieldCorrection** function to compare the effects before and after FFC.
4. Choose to either write coefficients (including **FFCFrameCount**) to device storage, or save parameters to file for future reuse.

See details in “MARS 10GigE Cameras User Manual” 8.4.9 section.

**For Scenario Type1 cameras:**

1. Put the lens aim to white paper or plane light source before acquiring images.
2. Click **Brightfield Images**, Optionally **Darkfield Images**.
3. With **FFCExpectedGrayValueEnabled** enabled (checkbox selected): Modify the **FFCExpectedGray** value and re-calculate - displayed images reflect the set value. Without checkbox selected: Input **FFCExpectedGray** value will not effect.

**For Scenario Type2 cameras:**

1. Put the lens aim to white paper or plane light source before acquiring images.
2. Click **Brightfield Images**, Optionally **Darkfield Images** (MARS-3140-3GC-P-M02 only).
3. Modify the **FFCExpectedGray** value and re-calculate - displayed images reflect the set value (MARS-3140-3GC-P-M02 only).

**For Scenario Type3 cameras:**

1. Set the **FFCCoefficient** (Not necessary).
2. Set the **FFCAccuracy** (**PixelLevel**: pixel-level correction. **BlockLevel**: block-level correction).
3. Begin Acquisition and calculation cannot be started while acquisition is stopped.
4. Click **Calculate.** Upon successful completion, the **FlatFieldCorrection** will automatically enable.
5. With **FFCExpectedGrayValueEnabled** enabled (checkbox selected): Modify the **FFCExpectedGray** value and re-calculate - displayed images reflect the set value. Without checkbox selected: Input **FFCExpectedGray** value will not effect.

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- **Acquisition of Brightfield Images**

1. When in **Stop Acquisition** status, clicking the **Brightfield Images** will display images in the GalaxyView acquisition interface.
2. When in **Start Acquisition** status, clicking the **Brightfield Images** will directly complete the bright field image acquisition.
3. The number of bright field images is related to the **FFCFrameCount**, for example, if you set the **FFCFrameCount** as 4, then 4 images will be acquired and applied to FFC calculation.
4. If the brightness of the acquired bright field images is below 20, then a warning massage will appear indicating it may affect FFC effectiveness. In this case, it's recommended to adjust brightness 20-250 and reacquire.
5. If the brightness of the acquired bright field images is exceed 250, then a warning massage will appear indicating it may affect FFC effectiveness. In this case, it's recommended to adjust brightness 20-250 and reacquire.

**Instruction:**

1. The bigger you set of the **FFCFrameCount**, the longer it will take to acquire images.
2. If a color camera is working without white balance, then you will get a white balance image after FFC. (Scenario Type1 and Scenario Type2 MARS-3140-3GC-P only)

- **Execute FFC**

1. Only after **Brightfield Images** acquisition is finished, can **Calculate** be enabled.
2. Click **Calculate** and calculate the FFC coefficient, then applies it to the device immediately, if not, you have to execute FFC again in case of power off caused data loss.
3. Check **FFCExpectedGrayValueEnabled** to see the effect before and after FFC (**FFCExpectedGrayValueEnabled** will take effect when FFC is finished).

- **Device: Load/Save**

1. Whether to execute **Device: Load/Save**, the FFC function will be enabled by default, so once you succeed in loading from device, FFC will take effect immediately.
2. The device will save the UserSet Setting as UserSet0 when saving into device (This feature is not applicable to scenario Type3 cameras).

- **PC: Load/Save**

1. Whether to execute **PC: Load/Save**,, the FFC function will be enabled by default, so once you succeed in loading from file, FFC will take effect immediately.
2. When load or save FFC file, the default path is (Under the installation path): *\GalaxySDK\Demo\Win64\resource\gxplugins\FlatFieldCorrection

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**Instruction:** Only .ffc file can be permitted while load from file.

- **Situations Requiring FFC Recalculation After Calibration**

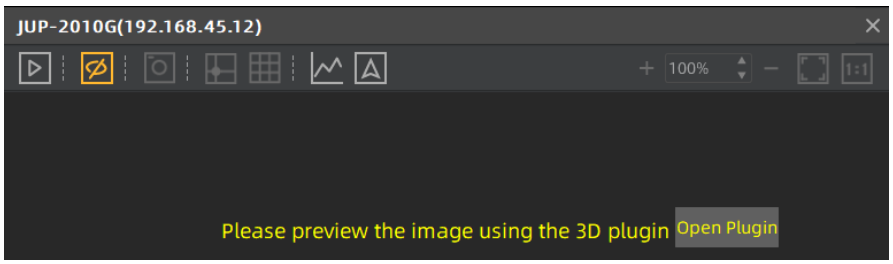
| User Action | Scenario Type1 | Scenario Type2 | Scenario Type3 |
|---|---|---|---|
| Lens replacement | Recalibration required | Recalibration required | Recalibration required |
| Gain adjustment | Recalibration required | Recalibration required | Recalibration required |
| ROI adjustment | Recalibration required | Recalibration required | Recalibration not required |
| Binning adjustment | Recalibration required | Recalibration required | Recalibration not required |
| Decimation adjustment | Recalibration required | Recalibration required | Recalibration not required |
| ReverseX adjustment | Recalibration required | Recalibration required | Recalibration not required |
| ReverseY adjustment | Recalibration required | Recalibration required | Recalibration not required |

### 3.3.7. 3D Manager Plugin

This plugin is suitable for the JUP series.

#### 3.3.7.1. Open Method

Method1: When the camera is in 3D mode, there is a button "Open Plugin" in the preview window to open the plugin directly.




Method2: Menu Bar → Plugin → 3D Manager Plugin

#### 3.3.7.2. Interface

- Function overview

1) Save the image

2) Load the local image

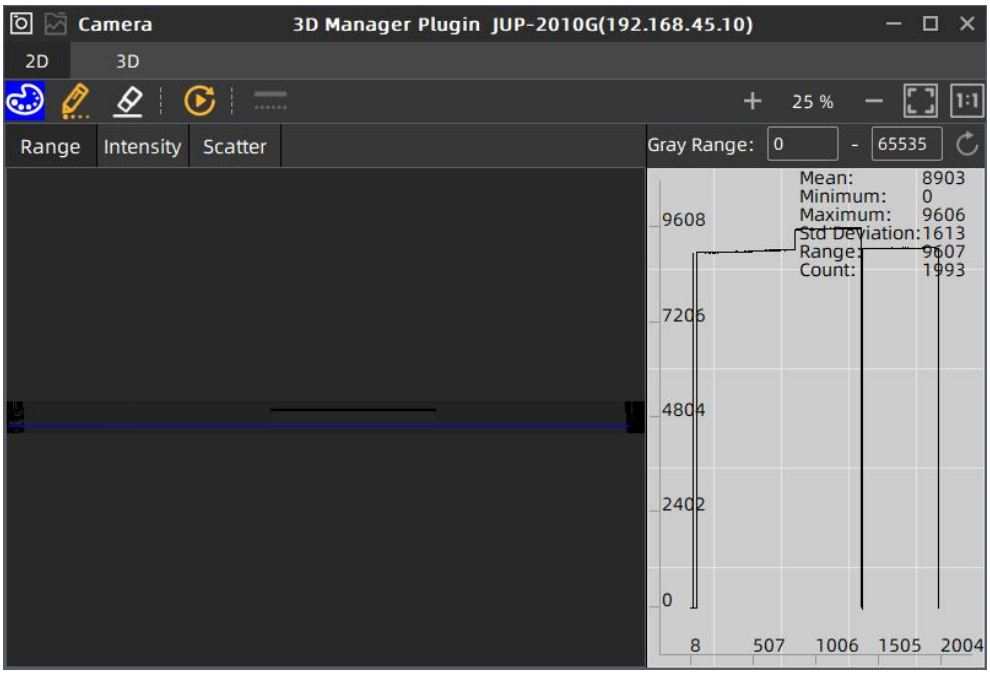
- 2D interface core functions

1) Use tabs to display Range, Intensity and Scatter

2) Each tab supports a standard horizontal or vertical line for the brush tool, which can be synchronized on the right to generate statistics for this line.

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3) Users can evaluate the extraction effect of the center of the laser line directly in the 2D laser line image.

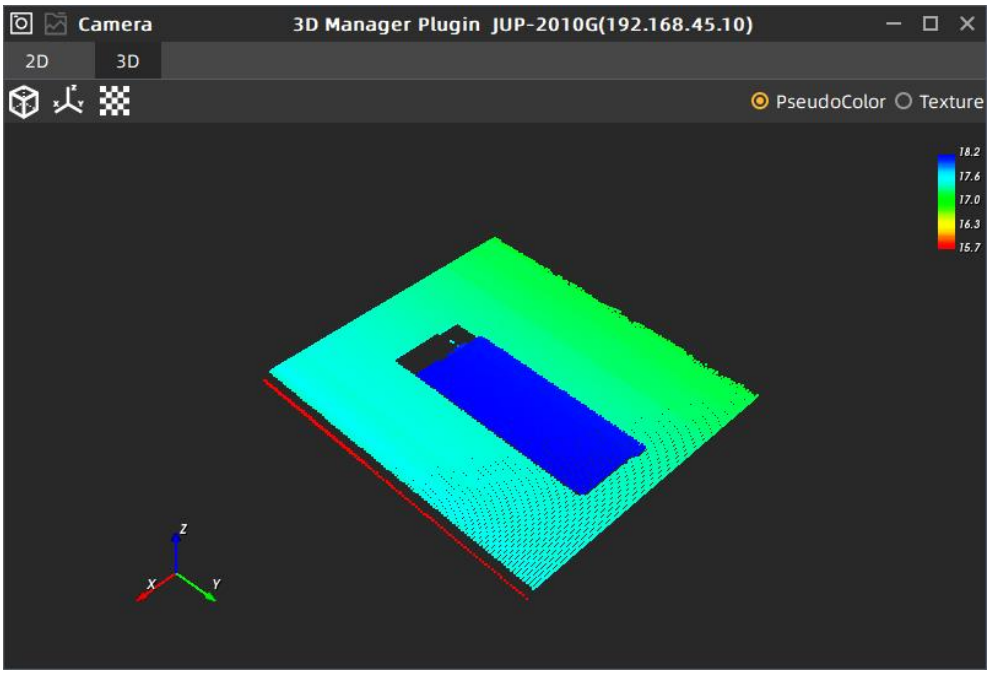



- 3D interface core functions

1) Generate point clouds

2) Pseudo-color color or texture rendering of the point cloud

3) Calibrate the file configuration




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3.3.7.3. User Guide

3.3.7.3.1. Adjust the laser line quality

Step 1: Connect the laser cable.

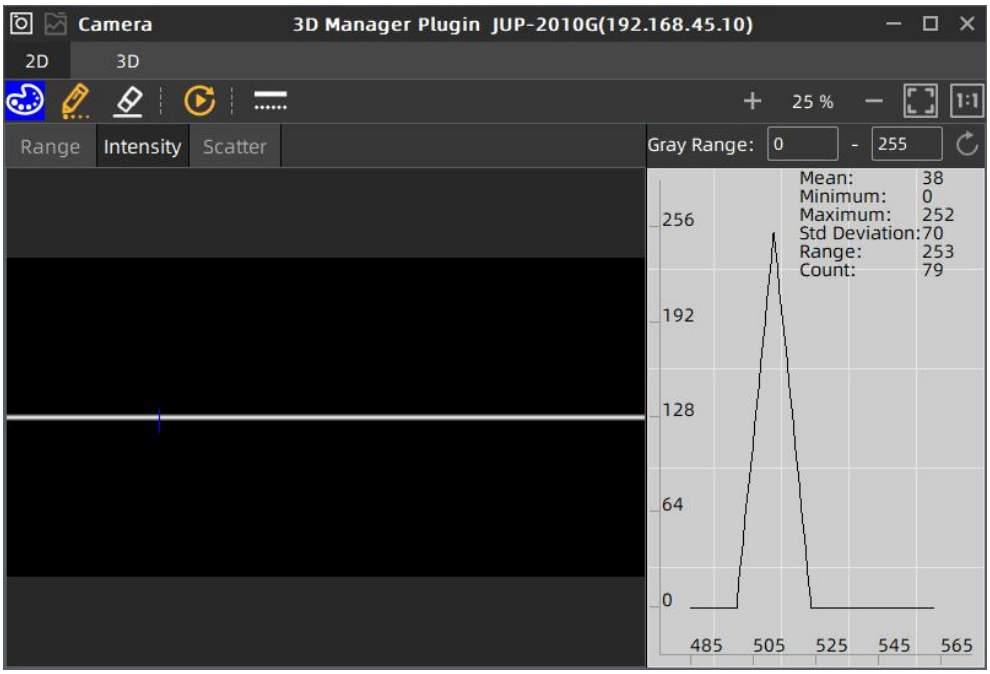
Step 2: Open the camera by GalaxyView and set the camera to 2D mode (DeviceScanType: Areascan).

Step 3: Turn on the laser and set the laser brightness (LightControllerSource: AlwaysOn or ExposureActive).

Step 4: Adjust the exposure time to a suitable value. ExposureTime is set to 100μs (according to the actual conditions).

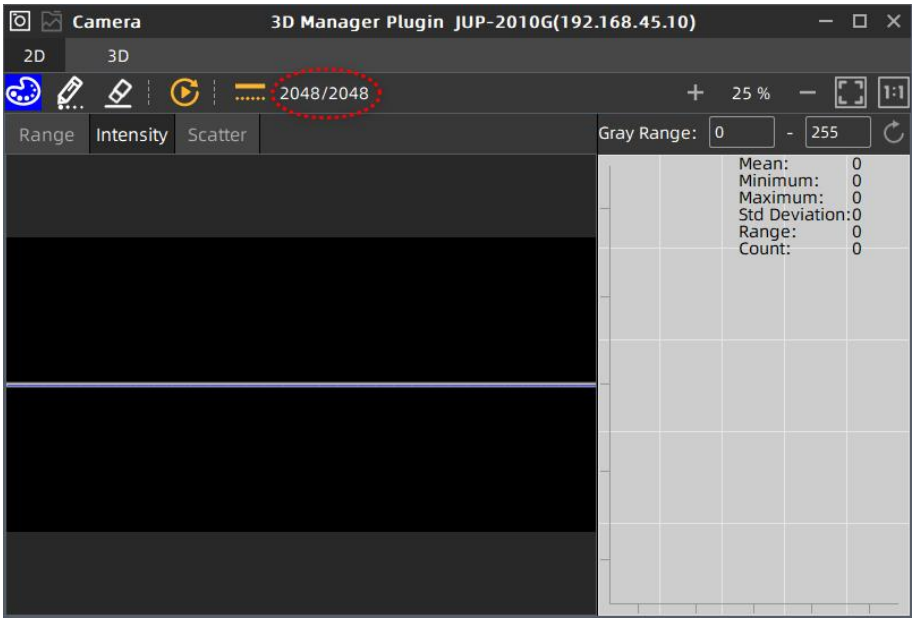
Step 5: Start the acquisition, then open the plugin and switch to the 2D image, you can see a bright laser line on the left display area.

Step 6: Use the brush tool to draw a vertical line on the display area of the left laser line, and on the right side, a statistical graph will be generated corresponding to the pixel points on this vertical line, to observe whether the peak is significant, and whether there is any data overflow.




Step 7: Evaluate the laser line quality with the laser line center point extraction evaluation function. After clicking the button, the number of successfully extracted points will be displayed next to the button, the closer the number of points is to the X resolution, the better the extraction quality. Users can adjust the brightness, the angle of the laser, the exposure time, the multi-slope mode (HDR) and the parameters of the extraction algorithm to improve the extraction result. Virtual dots will be generated on the left laser line display area to represent the center of the extracted laser line.

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3.3.7.3.2. View Range/Intensity/Scatter

Step 1: Set the camera to 3D mode. (DeviceScanType: Linescan3D).

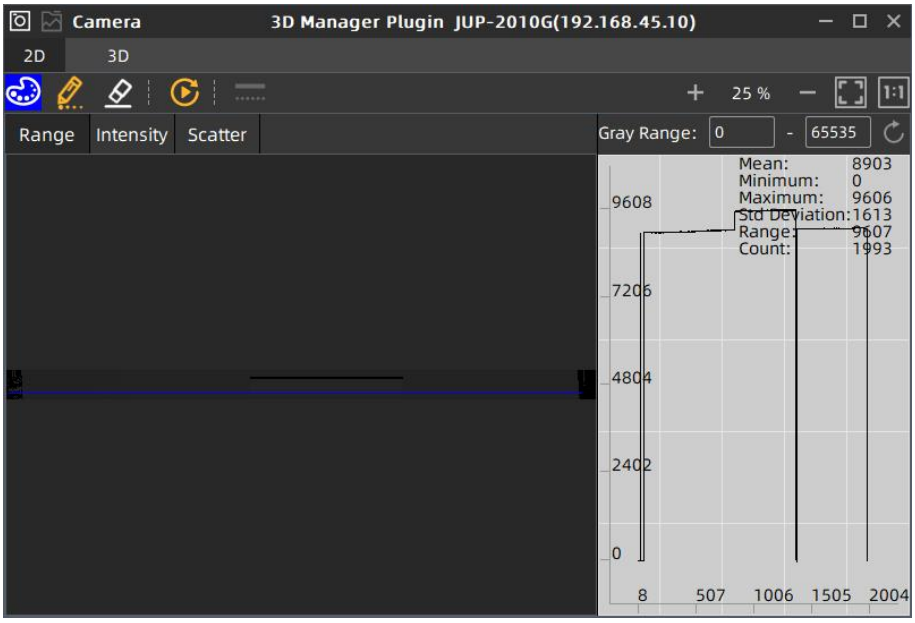
Step 2: Configure the component. In this example, all components will be enabled, ComponentSelector selects all components in turn, and then ComponentEnable is set to True.

Step 3: Configure the Height of the 3D image

Step 4: Start the acquisition, open the 3D plugin and switch to the 2D image:

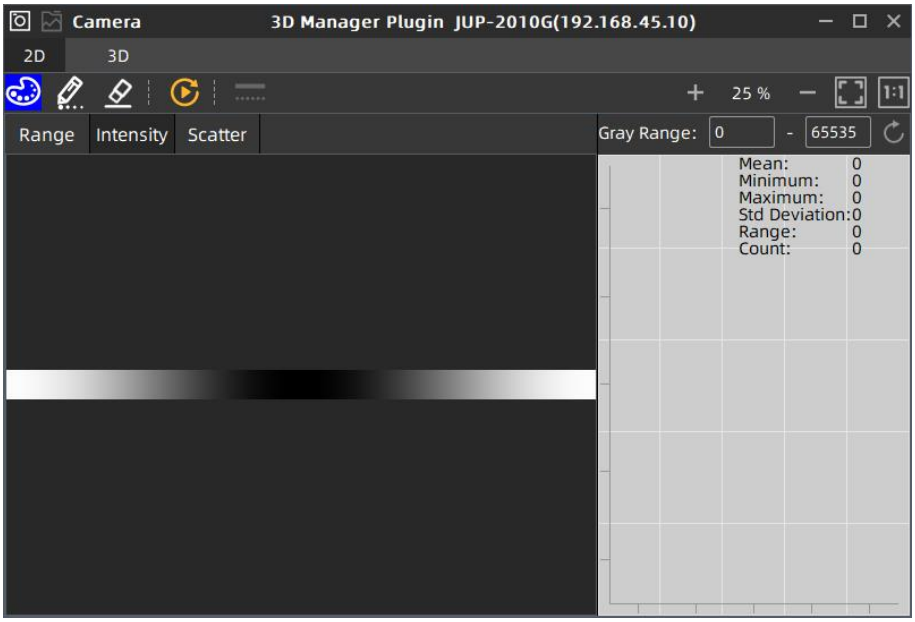
- **View Range**

Use the brush tool to draw a horizontal line on the Range, the horizontal line represents a profile, and on the right side, a statistical information will be generated corresponding to this profile.

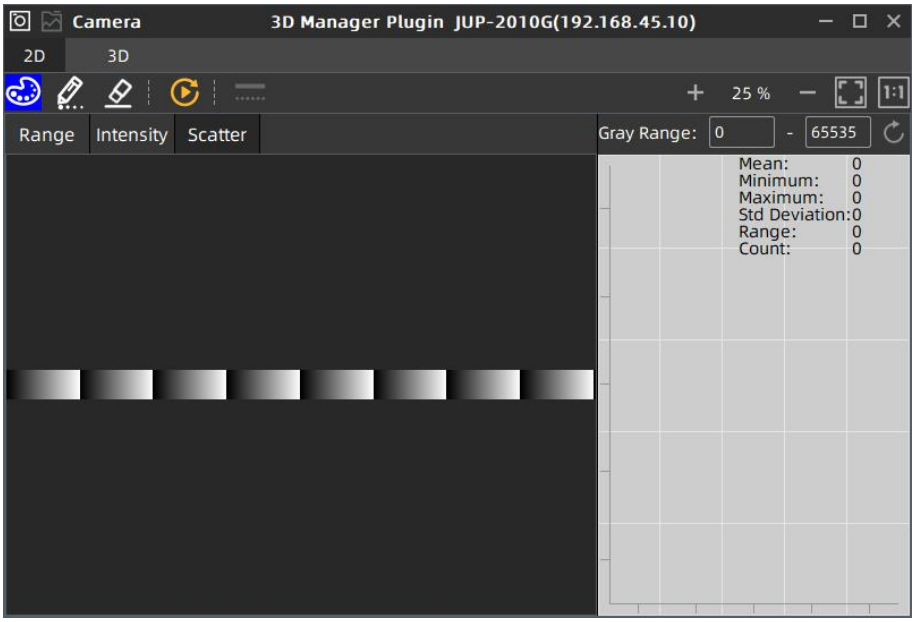



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- View Intensity




- View Scatter

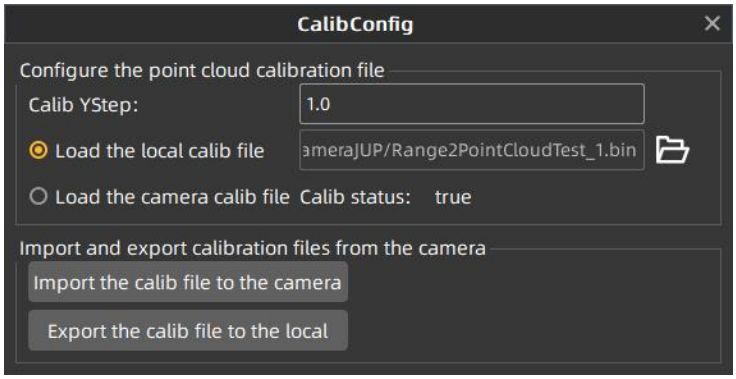



3.3.7.3.3. Generate point clouds

[Prerequisite: the camera is still in 3D mode, and the camera outputs Range and Intensity].

1) Switch the 3D plugin to the 3D page, and click the calibration configuration button "  " to configure the calibration file.

The calibration file can choose to load the local calibration file or use the calibration file stored in the camera.




Calib YStep represents the distance between adjacent lines in the Y-direction, the calculation formula is as follows:

$$dY=\frac{\text{Platform movement speed}}{\text{Acquisition line rate}}$$

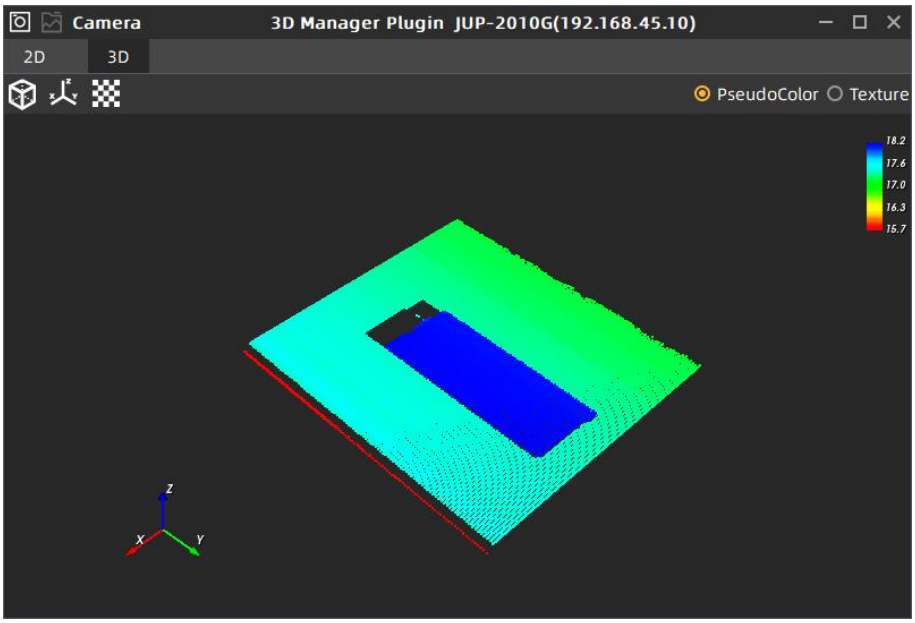
For example, assuming that the user system uniformly measures the length in millimeters, the units of tooth width, tooth height, and translation table movement distance used in the calibration process also need to be converted to millimeters. If the movement speed of the translation table is 10 millimeters per second, and the camera captures a line rate of 1000 lines per second, substitute it into the above formula will obtain:

$$dY=\frac{10\ mm/s}{1000\ Line/s}=0.01\ mm/Line$$

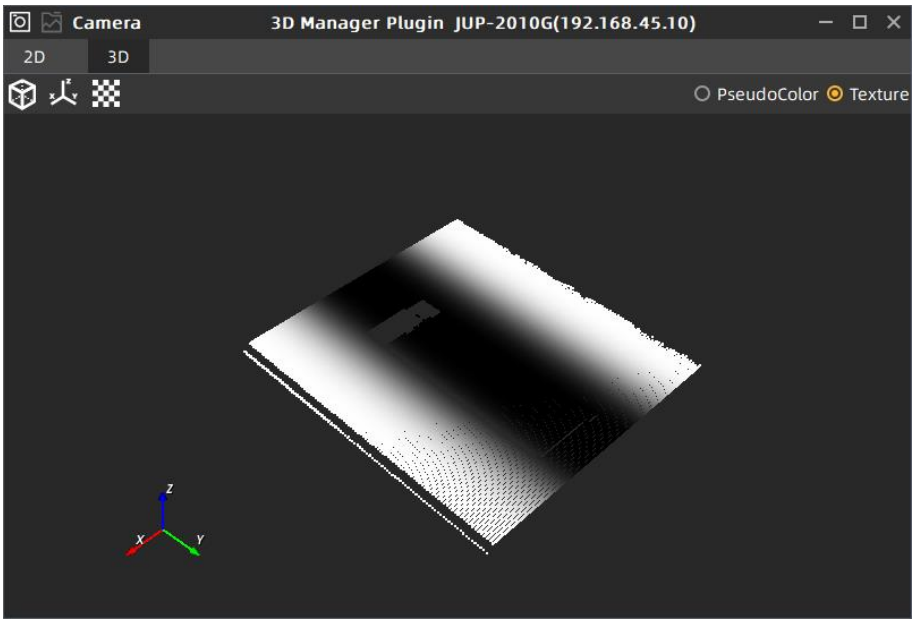
So, the value of Calib YStep should to be set as 0.01.

**Instruction:** When the dY's value is inaccurate, the generated point cloud will undergo compression or stretching deformation compared to the actual shape.

2) Click the build button " " "to generate the point clouds (default is pseudo-color image)



3) Choose "pseudo-color" or "texture" 

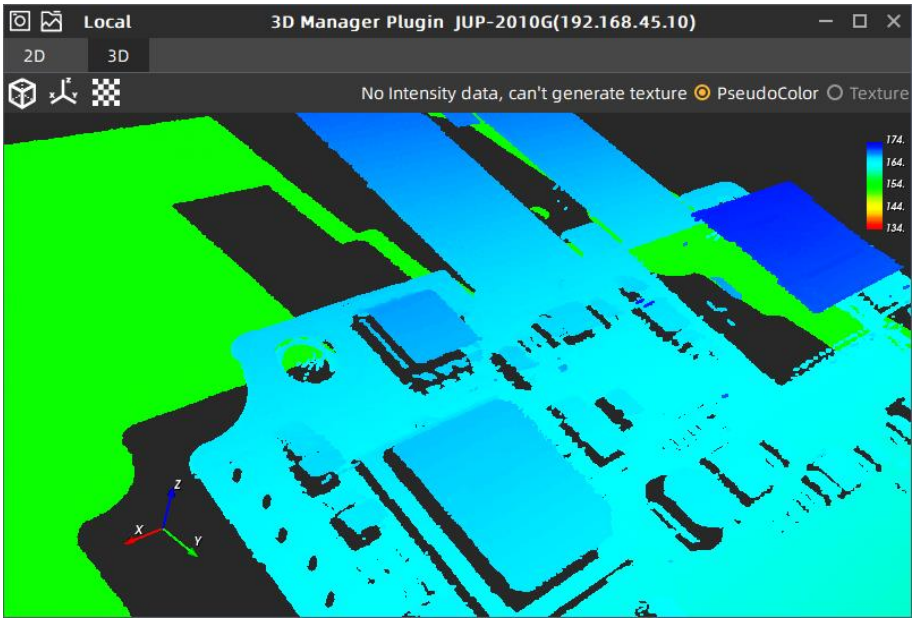




3.3.7.3.4. Preview local image

Step 1: Click the "Camera" button on the menu bar of the plugin. The button will change to "Local" after clicking.




Step 2: Click "Open Image" button, select the image you want to load (.tif, .ply or .pcd), in this example, load a local .ply image for demo (The local .ply file doesn't have the corresponding Intensity information, so it can't be used for texture rendering).




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3.3.7.3.5. Save image

Users can save the current image by clicking the Save Image button on the top left corner of the plugin title bar.




## 3.4. Settings

Including **General Settings**, **Save Image Settings** and **Internet Settings**.

### 3.4.1. General Settings

Here, you can change User Level (Beginner, Expert or Guru), Device list update automatically, Property list update automatically and The number of image buffer caches.

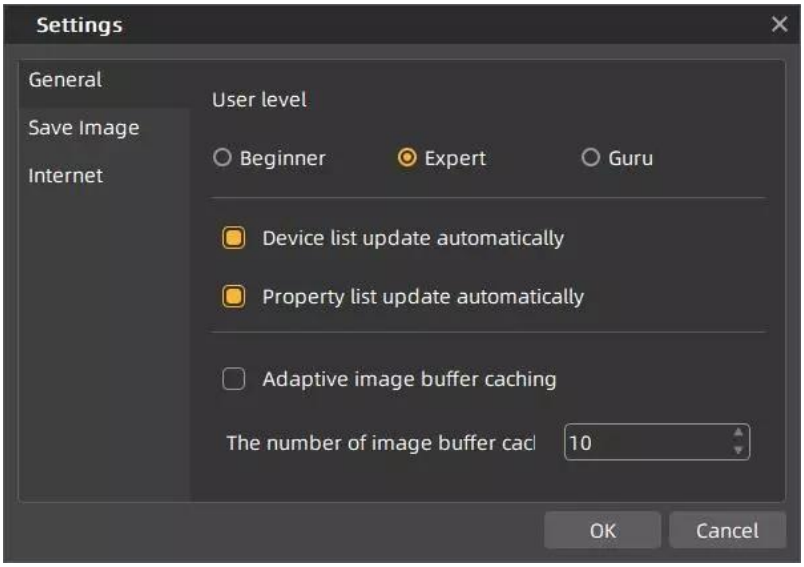



Figure 3-11 General Settings

- **User Level**

There are 3 options for you: Beginner (Default), Expert and Guru. The difference between them are the inside parameters, the higher the level is, the more parameters they have.

- **Device list update automatically**

With this option, GigE and USB devices will update automatically every 2s, otherwise you have to update the device list manually.

- **Property list update automatically**

With this option, parameters with Polling features will update and show automatically every fixed time, otherwise, you have to update and show the property list manually.

- **Adaptive image buffer caching**

When “Adaptive image buffer caching” is selected, the software controls the number of image buffers based on camera data. Conversely, users manually input the number of image buffer caches.

- **The number of image buffer caches**

The default value is 10, and this operation will be globally used to all connected devices.

**Instruction:** All settings above can be globally used to every cameras.

### 3.4.2. Save Image Settings

Here you can set the save path, save the images with/without editing effects, choose the picture formats and set the file name or format.

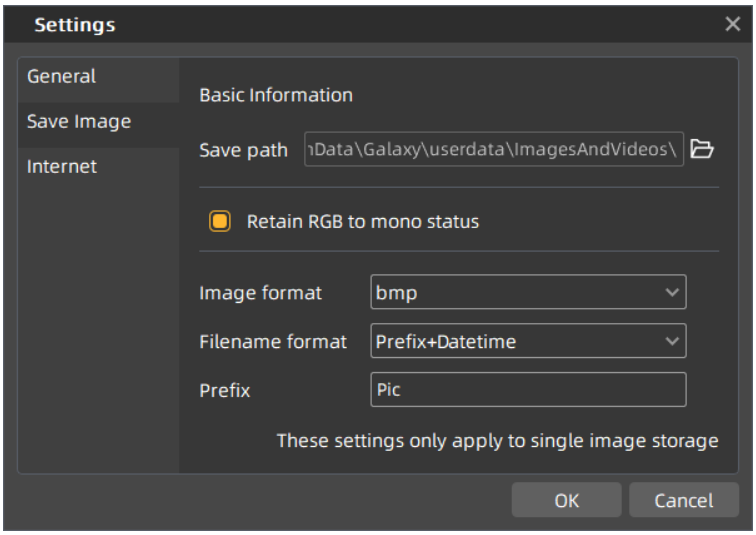



Figure 3-12 Save image settings

- **Save path**

Users can customize the file saving path by themselves, the default save path is:

C:\ProgramData\Galaxy\userdata\ImagesAndVideos

- **Retain RGB to mono status**

This function can maintain the adjusted image (Turn Left, Turn Right, Mirror, RGB to Mono) status, but only valid for bmp, jpg, png, tiff formats, raw formats can not be affected by this option.

- **Image format**

Five formats are available: bmp, jpg, png, tiff, raw.

- **Filename format**

There are **Prefix+Index** and **Prefix+Datetime** two options, prefix can be user-defined.

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### 3.4.3. Internet Settings

Including **GigE camera open mode**, **Adaptive network packet length**, **Device enumeration method** three parts.

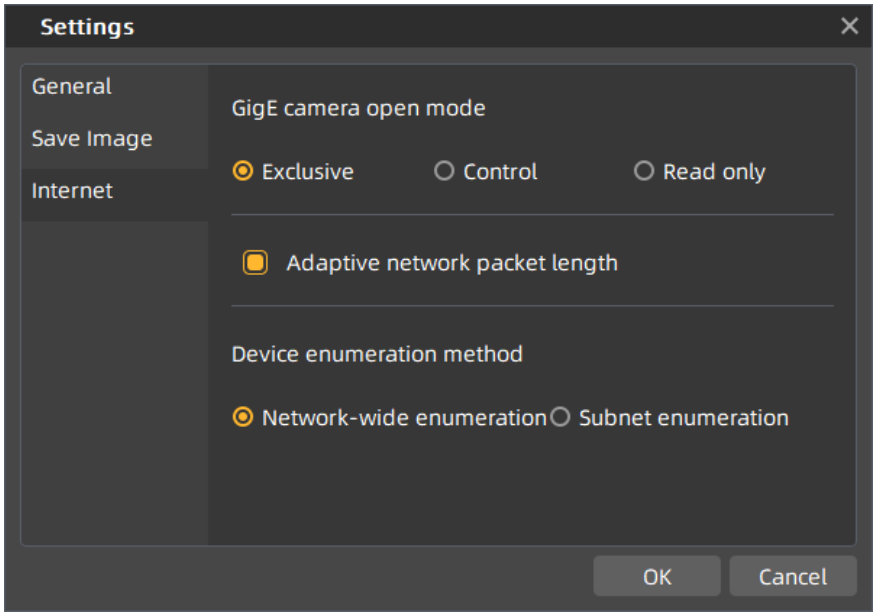



Figure 3-13 Internet Settings

- **GigE camera open mode**

There are three modes available: **Exclusive**, **Control** (Default), **Read only**.

| GigE camera open mode | Description |
|---|---|
| Exclusive | Forbid other processes to open the camera |
| Control | Allow other processes to open the camera in Read only mode |
| Read only | There is no influence for other processes to open the camera |

- **Adaptive network packet length**

With this option, the value of the PacketSize will be set automatically according to the network conditions.

- **Device enumeration method**

Here are **Network-wide enumeration** and **Subnet enumeration** two options.

➢ Network-wide enumeration: Send a 255.255.255.255 destination address broadcast packet to the new host, by then you can search all cameras in the LAN no matter whether in the same network segment.

➢ Subnet enumeration: For instance, if your NIC IP address is 192.168.40.10, a 192.168.40.255 broadcast packet in subnet will be sent, so you can just search the cameras in the same subnet.

**Instruction:** All settings above are globally effective to every cameras.

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## 3.5. Tools

### 3.5.1. GxGigEIPConfig

You can set the IP Address and IP Configuration by **GxGigEIPConfig.**

- **Guidance**

Menu bar → Tools → GxGigEIPConfig, see Figure 3-14:

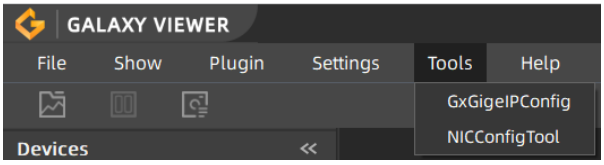



Figure 3-14

See GxGigEIPConfig interface as Figure 3-15:

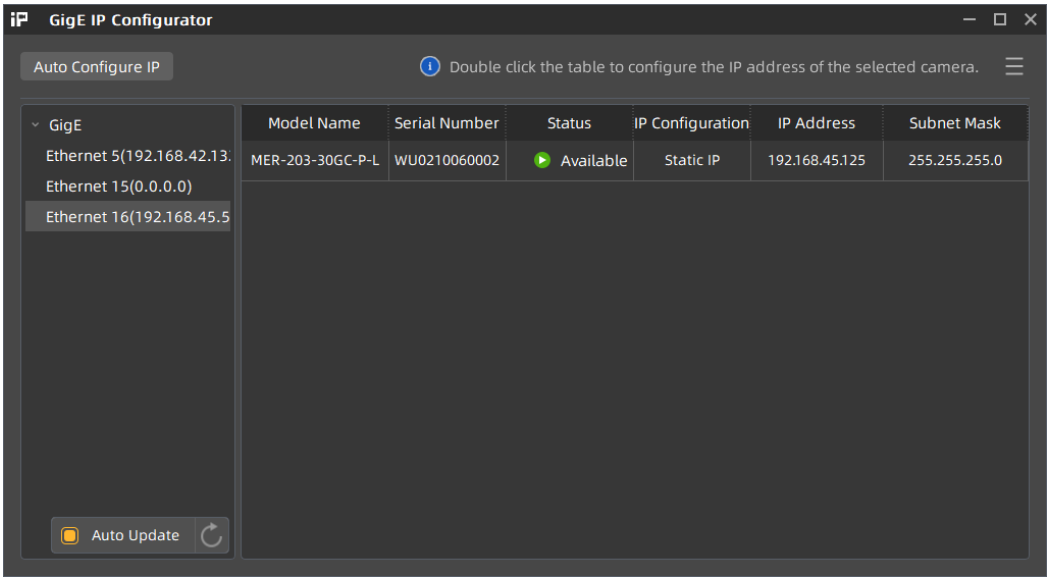



Figure 3-15 GxGigEIPConfig interface

The left side of the interface is device list, showing all searched GigE (Gigabit Ethernet)

1. When GigE is selected, all GigE cameras will be shown in the right side.
2. When selecting a certain GigE, the right side will only show the corresponding GigE cameras, and if you want to change the Ethernet attributes, just right click it.

**Instruction:**

1. **Auto Update** will enumerate devices automatically in every 2s, you can also click " " to update devices manually.
2. The essential information are listed in the right side, click " " in the top right-hand corner to set.

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- **Status and operations**

Status: Available, Read Only, Occupy, Unreachable, Unknown.

Operations: Auto Configure IP, Modify IP Address, Release Occupation, Reset Device.

| Icon | Status | Description | Operation |
|---|---|---|---|
| | Available | When the camera is not opened in Exclusive or Control mode by other processes. And the camera IP can be modified manually | 1. Modify IP Address<br>2. Reset Device |
| | Read Only | When the camera is opened in Control mode by other processes. By this time, the Auto Configure IP and Modify IP Address will be forbidden | 1. Release Occupation<br>2. Reset Device |
| | Occupy | When the camera is opened in Exclusive mode by other processes. By this time, the Auto Configure IP and Modify IP Address will be forbidden | 1. Release Occupation<br>2. Reset Device |
| | Unreachable | Three conditions when in Unreachable status:<br>1. The current camera IP is same as other camera IP<br>2. The current camera IP is same as network card IP<br>3. The current camera IP and the connected network card are not in the same subnet | 1. Modify IP Address<br>2. Auto Configure IP |
| | Unknown | Cannot get the current accessible state for some reason | 1. Modify IP Address |

Table 3-4 Device status description

Valid IP: Non LLA address, non 0.0.0.0.

- **Auto Configure IP**

By this function, all unreachable camera's IP will be changed to the effective IP, which has the same network segment as the PC network port.

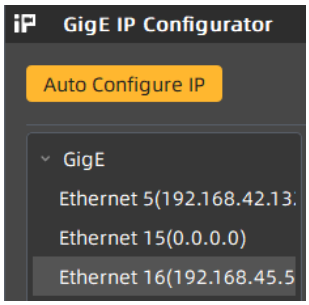



Figure 3-16

**Instruction:** If the host network card IP is invalid, then after Auto Configure IP operation, it will be valid.

- **Modify IP Address**

Users can double click the row of the camera in the list, the window as Figure 3-17:

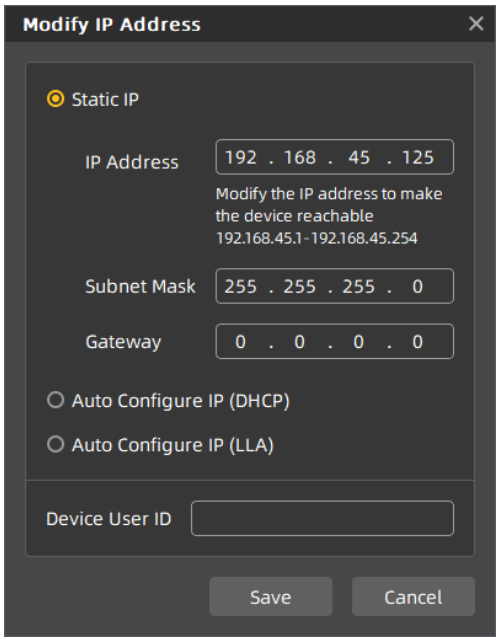



Figure 3-17 Modify IP Address

The default setting is **Static IP**, and in here, users can set **IP Address**, **Subnet Mask** and **Gateway**.

**Instruction:** It is restrict to set the IP as: class D (224~239), Class E (240~254), IP start with 127 or 255. There will be an error prompt if you set the restrict IP or wrong format IP, in this case, **Save** is not available. See Figure 3-18:

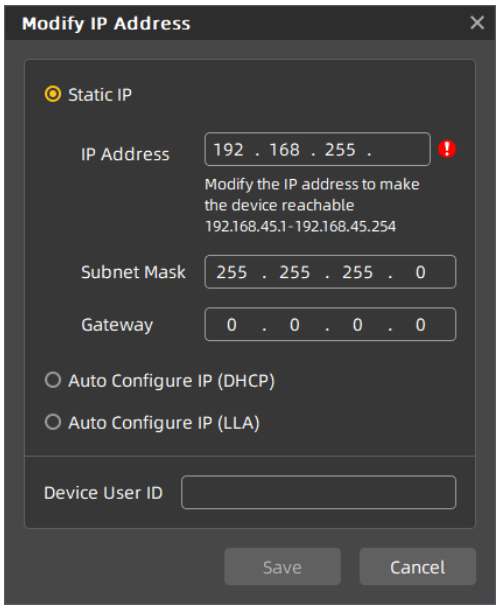



Figure 3-18 IP address format check

Users can set Static IP, DHCP or LLA according to their needs.

| IP Configuring Type | Description |
|---|---|
| Static IP | Default configuration, which is saved in the camera Flash sensor and cannot be changed while power on/off |
| DHCP | When start with DHCP, please ensure that DHCP server is configured in your network environment, otherwise, the camera will restart in LLA address after waiting DHCP server assign IP overtime. But when DHCP server occurs, the camera IP will switch to DHCP assigned address, the default factory configuration of the camera is DHCP |
| LLA | Link-local address, which is used for local network communication, not forwarded through routing |

Optional operation: Modify Device User ID, the maximum length of user-defined name is 16 characters.

Release Occupation: The camera heart beat time is 5min by default (VS develop, in debugging status). If users forcibly exits the process without closing the camera, then the camera cannot be resetting immediately, unless the 5min heart beat time is over, and by then, the camera is in Occupy or Unreachable status. Users can double click the row of the camera in the list to release the camera and open the camera again.

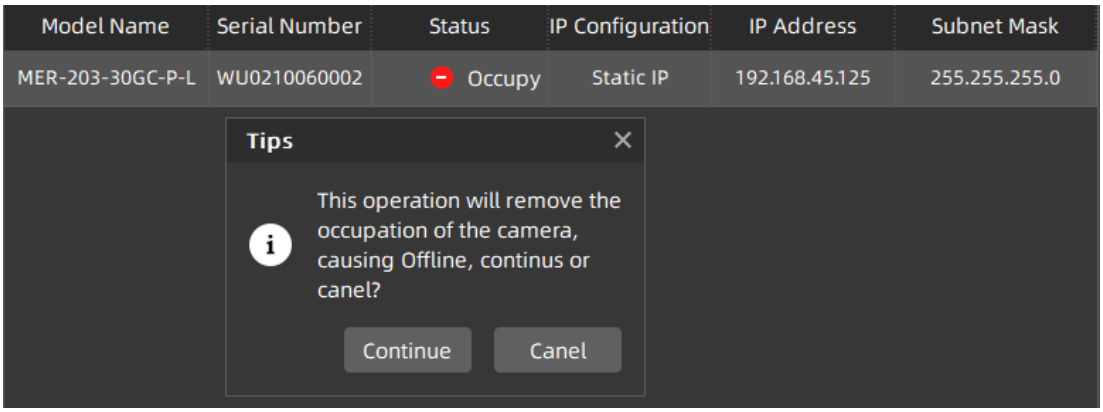



Figure 3-19

Reset Device: While users need to reload the camera and the device is not convenient to power off, then right click the row of the camera in the list and click Reset Device button.

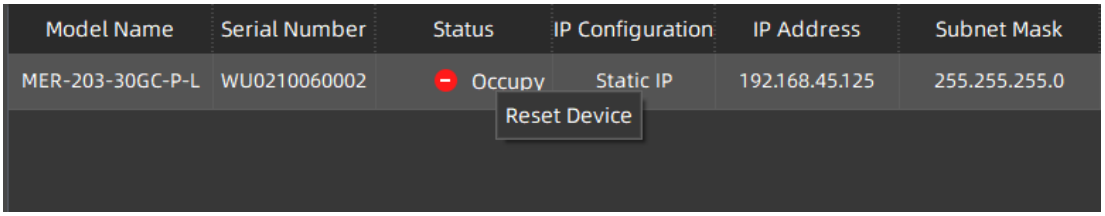



Figure 3-20

**Instruction:**

1. Be careful to use **Release Occupation** and **Reset Device**, they will cause the camera off-line while acquiring.
2. Require camera itself supports **Release Occupation** and **Reset Device**.

### 3.5.2. GxLogViewer

- Boot Mode

Method1: Windows → GxLogViewer

Method2: Tool Bar (GalaxyView) → Tool → GxLogViewer

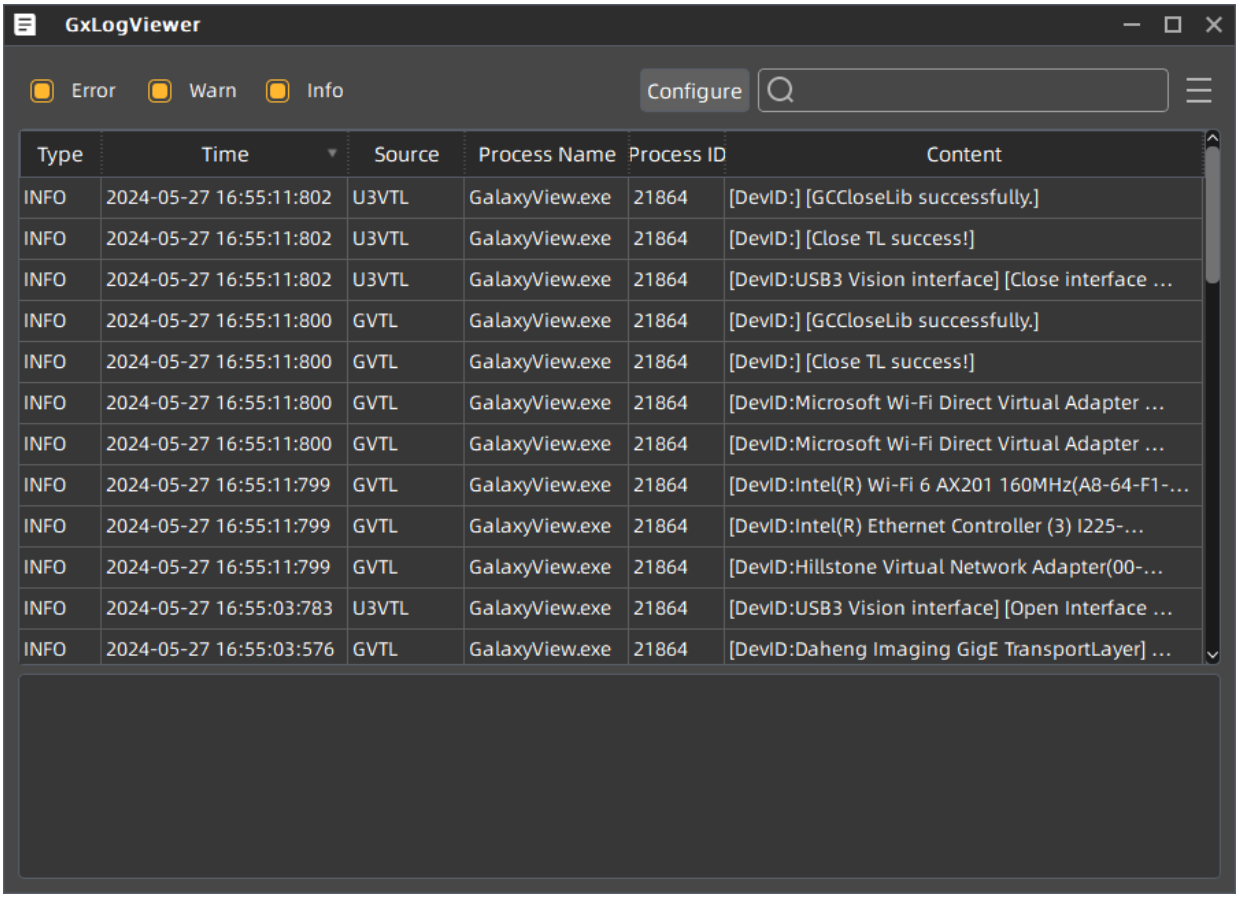



Figure 3-21 GxLogViewer window

Set log display content: Each log information includes Type, Time, Source, Process Name, Process ID, and Content. The content displayed in the log information can be set in the upper right corner .

View log types: There are Error, Warn, Info three types. You can determine whether a certain type of log is displayed by checking the log type in the upper left corner.

Query log: Input keywords and press enter to query. Only the content in the log can be entered as keywords for querying.

Log configuration: Users can set and check the log by clicking Configure.

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- LogViewConfig

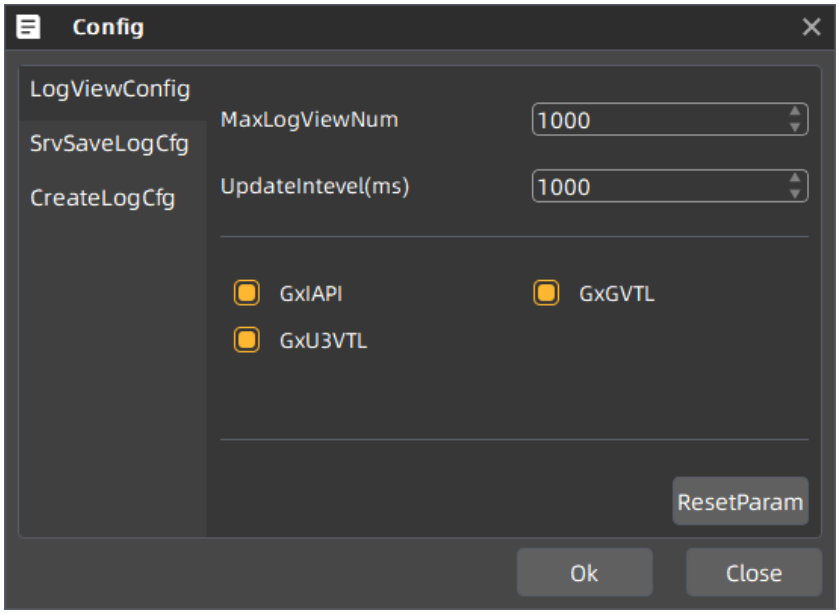



Figure 3-22 LogViewConfig

**MaxLogViewNum:** Set the number of log displays, with a default value of 1000 and a range of 1-10000. Not displayed when exceeding the set value for MaxLogViewNum.

**UpdateIntevel (ms):** Set the interval for log updates, unit in ms, with a default value of 1000 and a range of 1-60000.

**Log Source:** Set the module that needs to display logs, all modules will be displayed by default, including: GxIAPI, GxGVTL, GxU3VTL. Unchecking will not display the module logs.

**ResetParam:** Restore the MaxLogViewNum, UpdateIntevel and Log Source to their default values.

- SrvSaveLogCfg

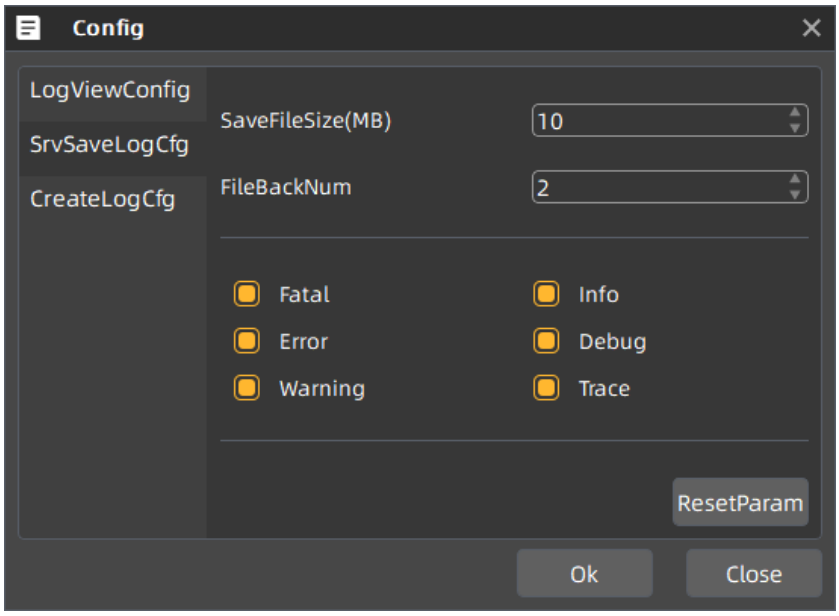



Figure 3-23 SrvSaveLogCfg

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**SaveFileSize (MB):** Set the file size of the single log, unit in MB, with a default value of 10 and a range of 1-1024. When the log file reaches the set value of SaveFileSize, a new log will be generated and saved, while the old log will be saved in the backup file.

**FileBackNum:** Set the number of backups for log files, with a default value of 2 and a range of 1-10. When the number of backup files reaches the set limit, the old backup files will be discarded and the newly generated backup files will be saved.

**Log Storage Type:** Set the type of log that needs to be saved in the log file. Save all types of logs by default, which includes Fatal, Error, Warning, Info, Debug and Trace.

**ResetParam:** Restore the SaveFileSize, FileBackNum and Log Storage Type to their default values.

- CreateLogCfg

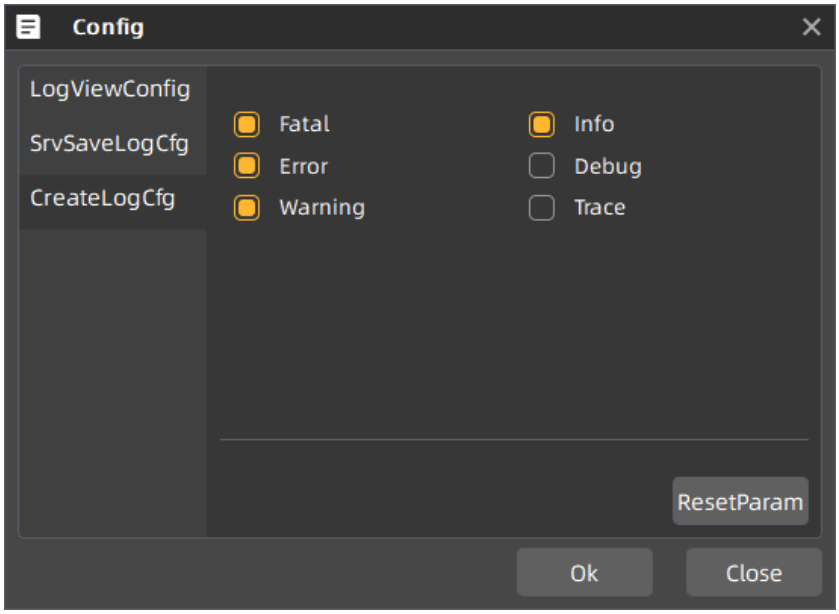



Figure 3-24 CreateLogCfg

**Log Generation Type:** Set the type of log to be generated. Generate logs of Fatal, Error, Warning and Info types by default. If you need to view Debug and Trace type logs, you need to check the corresponding options and view them in the Log file. The log file path is: C:\ProgramData\Galaxy\logdata\Log.log.

**ResetParam:** Restore the Log Generation Type to its default value.

### 3.5.3. GxUpgradeTool

- Boot Mode

Method1: Windows → GxUpgradeTool

Method2: Tool Bar (GalaxyView) → Tool → GxUpgradeTool

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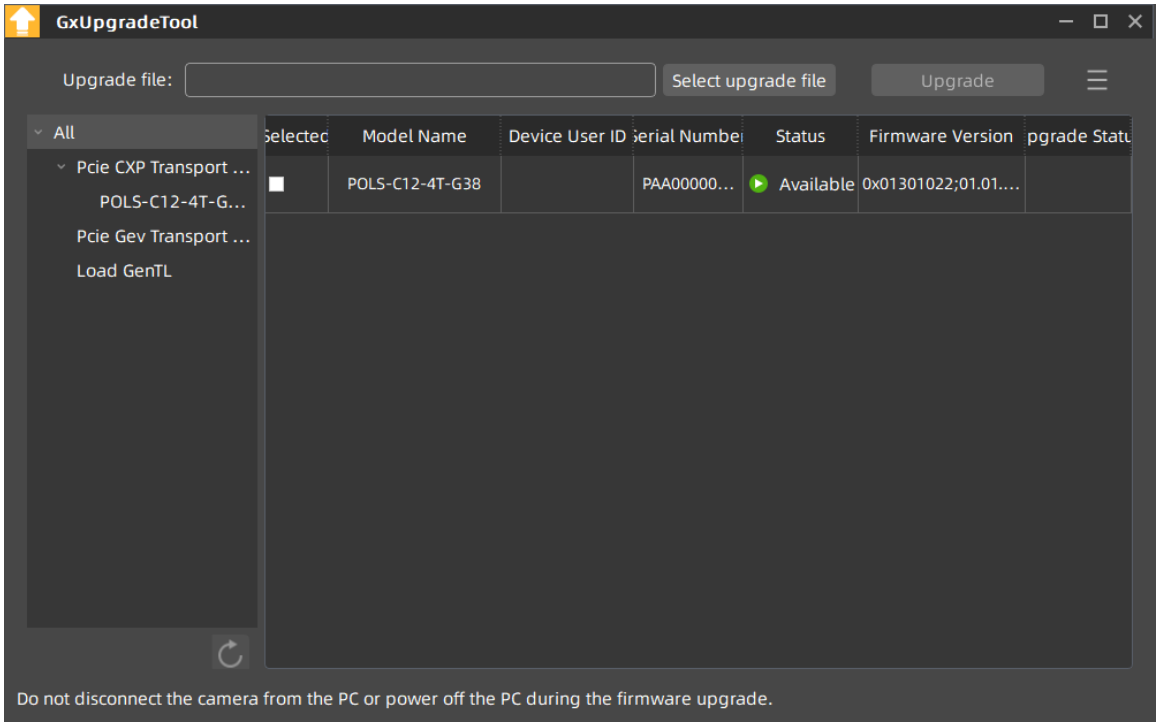



Figure 3-25 GxUpgradeTool

The left side in the upper figure is the device list, which displays the all devices searched.

When selecting Pcie Gev, all GigE cameras and NIC card will be displayed on the right side.

When selecting a network port, only the GigE cameras found under that port will be displayed on the right.

1) The "Auto Update" button below the device list automatically enumerates devices every 300 milliseconds by default, or you can click the " " button to actively update the device list.
2) The basic information of the camera is displayed on the right side of the tool, which can be set through the " " in the upper right corner of the tool.

- Status and Operation

The device status includes Available and Occupy.

| Icon | Status | Description |
|---|---|---|
| | Available | When the camera is not opened by other programs, the device status is "Available", indicating that the device can be upgraded |
| | Occupy | When the current camera is opened by another program, the device status is "Occupy", indicating that the device cannot be upgraded |

- When to Upgrade the Device

1) When the camera has new features that need to be upgraded.

2) When the dat file in the frame grabber does not match the dat file on the host, the frame grabber will enter factory mode, resulting in a red exclamation mark. At this time, it is necessary to upgrade with the matching programmed dat file. As shown in the following figure:

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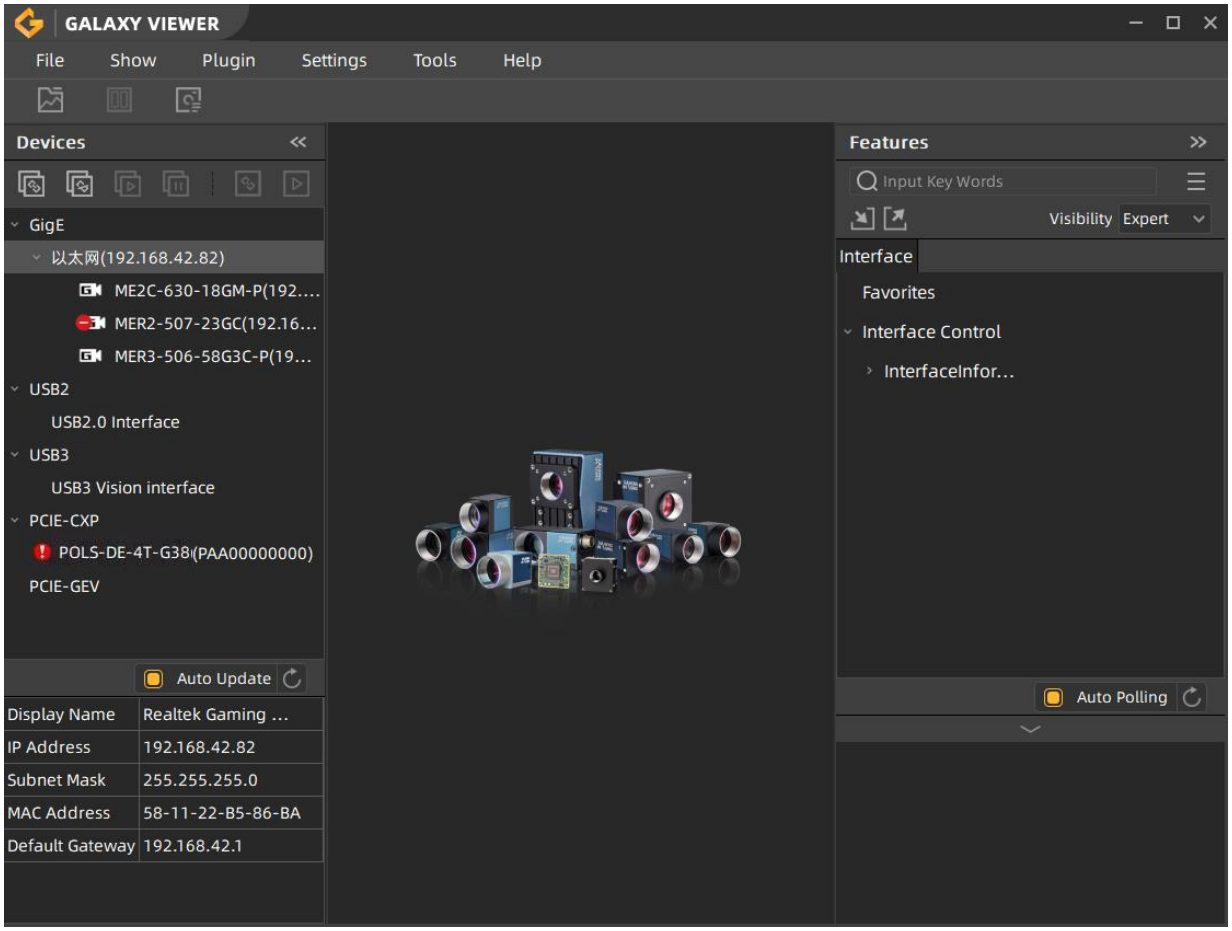
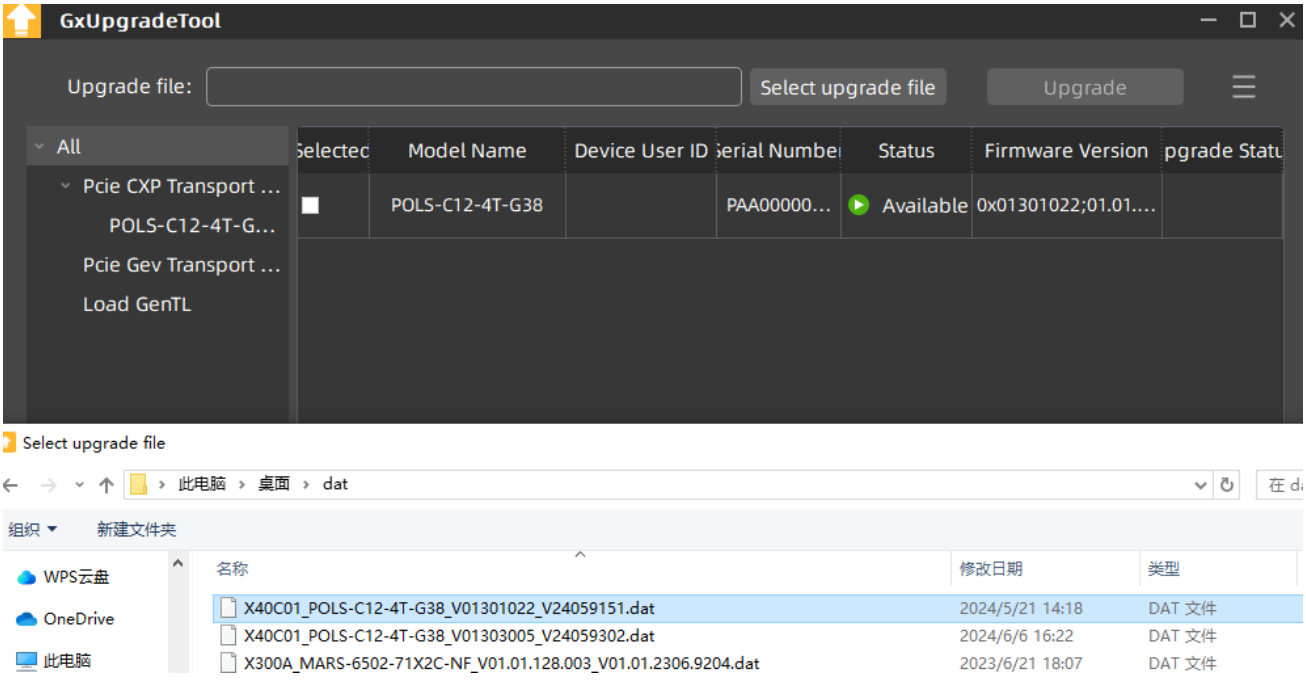



Figure 3-26 Frame grabber abnormality

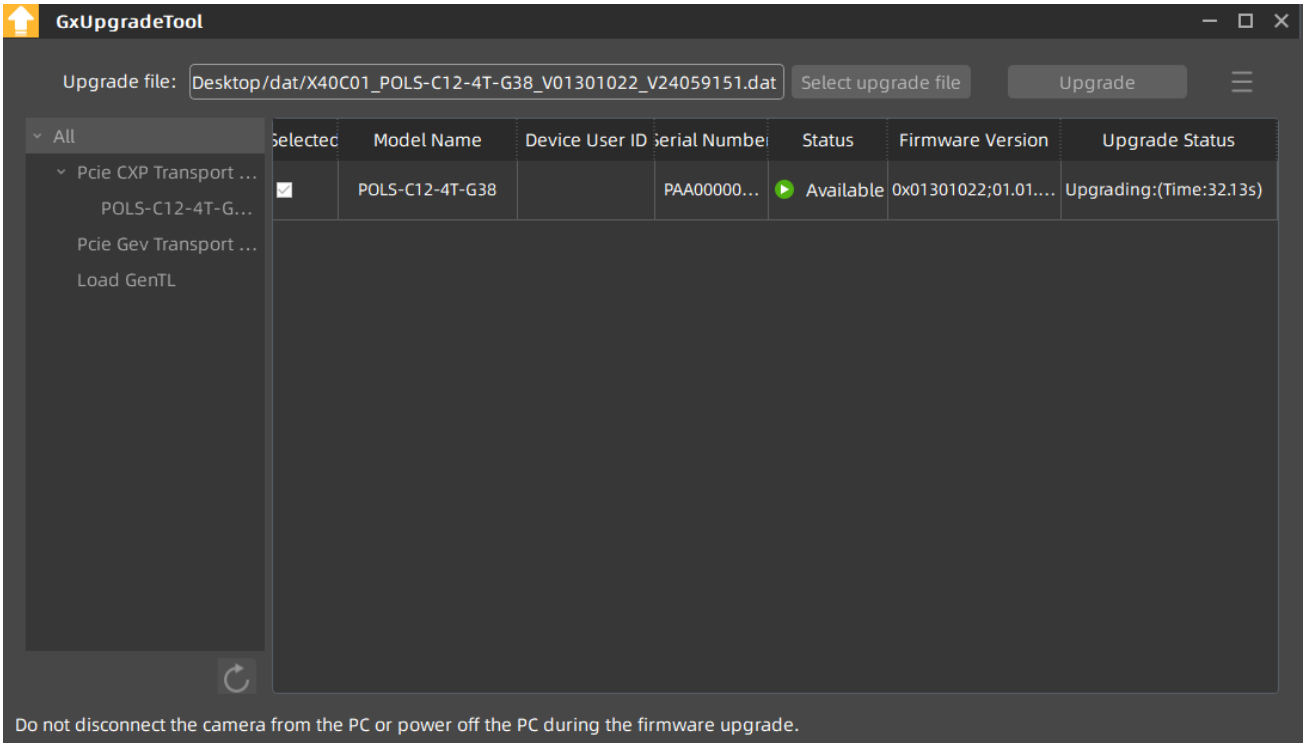
- Equipment Upgrade Method

1) Click "Select upgrade file" to open the folder where the dat file is located and select the corresponding dat file, see as the following figure:

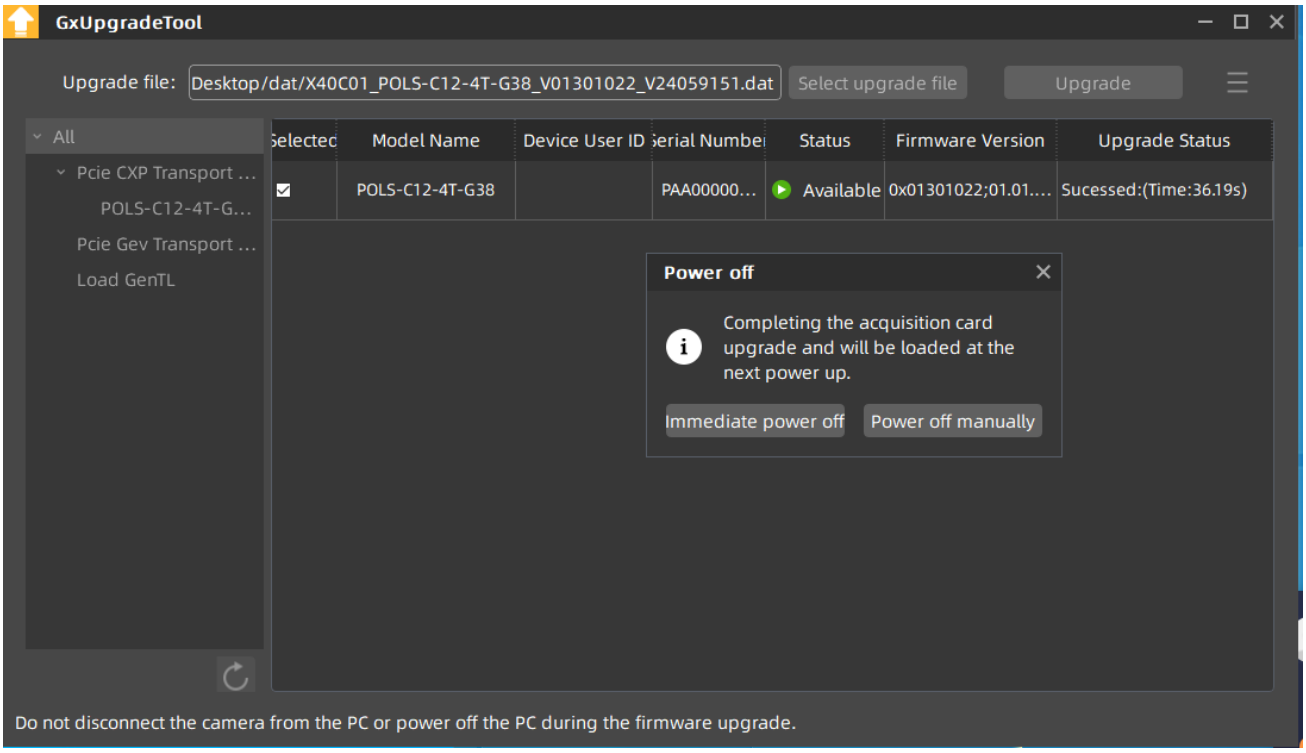



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2) After selecting dat file, click "Upgrade" to start upgrading:




3) After upgrading, there are two options you can choose: "Immediate power off", "Power off manually".



### 3.5.4. NICConfigTool

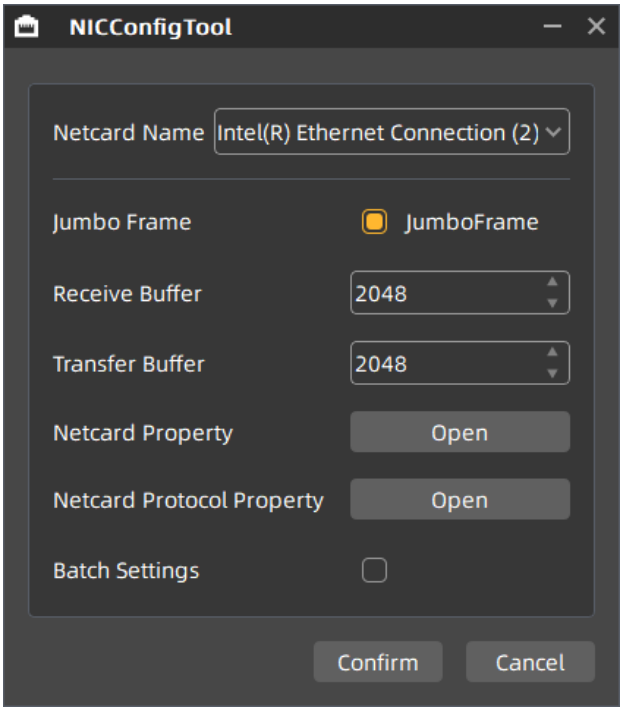



Figure 3-27 NICConfigTool

- **Guidance**

Method 1: Windows → NICConfigTool

Method 2: GalaxyView → Tool Bar → Tools → NICConfigTool

Method 3: Device List → Right click any NIC → NICConfigTool

Method 4: GxGigEIPConfig Device List → Right click any NIC → NICConfigTool

Select a NIC in the check box of the Netcard Name to set the parameters:

**Jumbo Frame:** Jumbo Frame can reduce CPU usage and improve data transmission efficiency when transmitting large frame length (Data latency will occur).

**Receive Buffer:** Copy the data to memory and set the NIC Receive Buffer value, the bigger the value is, the better the performance are, but more memories will be consumed.

**Transfer Buffer:** Copy the data to memory and set the NIC Transfer Buffer value, the bigger the value is, the better the performance are, but more memories will be consumed.

**Netcard Property:** Check and modify the configuration of the NIC, some device manufacturer set configurations cannot be changed by users.

**Netcard Protocol Property:** It is vital to set a valid IP for NIC and GigE Vision device communication, through this button, you can check or modify the contents of IPv4 check box.

**Batch Settings:** Allow whole settings (Jumbo Frame, Receive Buffer, Transfer Buffer) apply to all NIC devices.

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### 3.5.5. GxDriverManager

Users can view the current status of device drivers through this tool, as well as install/uninstall the drivers.

- **Guidance**

Find the installation package through the Start Menu, find GxDriverManager option and click it.

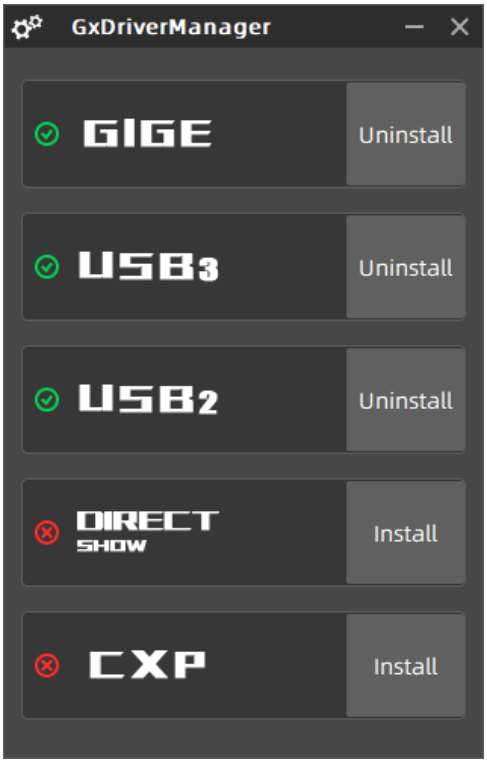



Figure 3-28 GxDriverManager tool

Installed successfully  Uninstalled

### 3.5.6. Action Command Window

Users can use this tool to batch configure multiple cameras to receive operation commands. Firstly, set the action command information in the **Action Command Parameters** area, which can be manually entered or click **Auto Config** to set default parameters. After selecting **Auto Config start acquisition** option, the cameras' trigger source will be set to Action and start acquisition by clicking **Auto Config** button. Clicking "**Send Action Command**" will trigger the camera to capture an image.

- **Boot Mode**

Tool Bar (GalaxyView) → Tool → Action Command Window

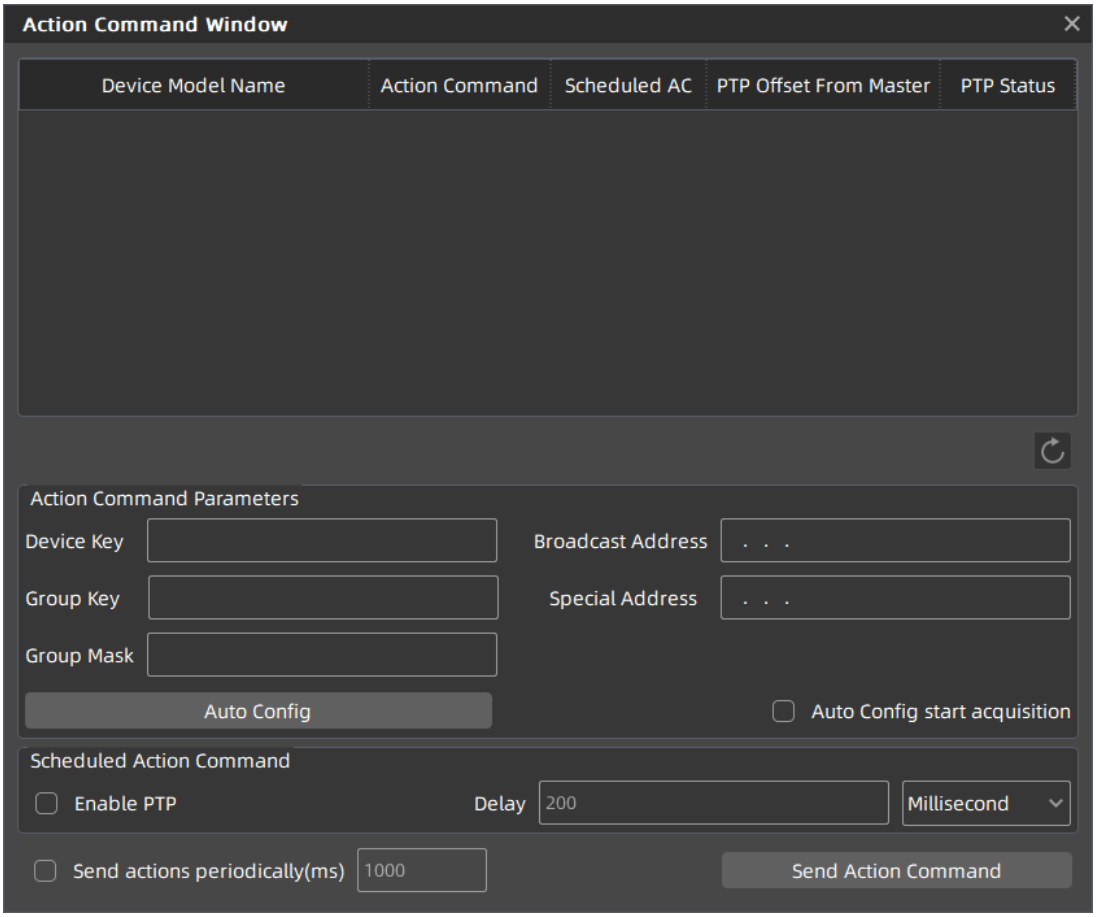



Figure 3-29 Action command window

The list at the top of the window displays all the opened cameras that are connected to your computer and support Action Command functionality. If there are no cameras in the list, please check in the device window if the connected cameras are turned on and if they support this feature. If you add or remove any camera that supports Action Command function and is already open in the device window, it will be immediately reflected in the Action Command Window.

**Action Command Parameters:** You can manually input the required parameters or choose Auto Config. If you want to write this setting to the camera, please select Auto Config start acquisition and click Auto Config. This operation will continue until the camera is disconnected.

**Broadcast Address:** This parameter informs the computer where the action command should be sent. The default is 255.255.255.255 (including all cameras connected to the computer). If you want to limit the number of cameras that can receive action commands, you can specify a single subnet or specify a single camera.

**Special Address:** This parameter informs the computer from which network adapter IP the actiont command should be sent. If not specified, each IP of all network adapters will send the current action command.

The following three parameters are used to configure the camera, all using 32bit hexadecimal notation.

**Device Key:** This parameter is used to authorize the execution of action commands on the camera, corresponding to the "ActionDeviceKey" property of the device.

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**Group Key:** This parameter is used to define a set of cameras on which action commands can be executed, corresponding to the "ActionGroupKey" property of the device.

**Group Mask:** This parameter allows users to filter out a camera subgroup from the group defined by the Group Key, corresponding to the "ActionGroupMask" property of the device.

**Auto Config start acquisition:** After selecting this option, click Auto Config will enable batch acquisition of cameras that support Action.

**Auto Config:** Click Auto Config button will set the Device Key, Group Key, and Group Mask values to the camera, while setting the Trigger Mode to On and Trigger Source to Action0.

**Send actions periodically (ms):** A timer will be activated every specific time (user input) to trigger the "Send Action Command" button to send actions.

### 3.5.7. GxPCDiagnostic

- **Boot Mode**

Method1: Windows → GxPCDiagnostic

Method2: Tool Bar (GalaxyView) → Tool → GxPCDiagnostic

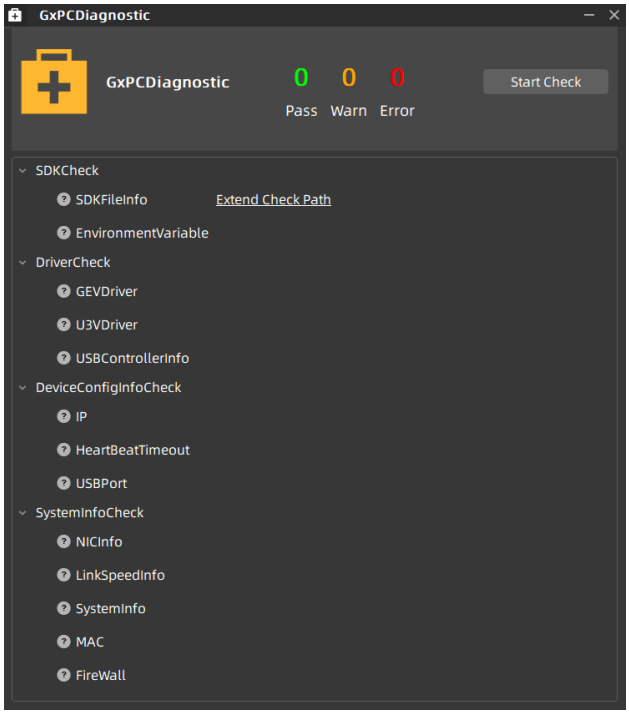



Figure 3-30 GxPCDiagnostic window

After launching the tool, you will see the UI as shown in the figure above. Clicking the "Start Check" button initiates a hardware inspection of the host. Before starting the check, you can click the "Extend Check Path" button. This button allows you to scan a custom directory to check for any SDK files whose versions do not match those included in the installation package.

**SDKFileInfo:** This check verifies if any files within the system paths have SDK versions inconsistent with those in the installation package. If mismatched files are found, the tool will prompt you to remove them. Click the "View" button next to the "SDKFileInfo" item to see detailed error information, as illustrated in Figure 3-31.

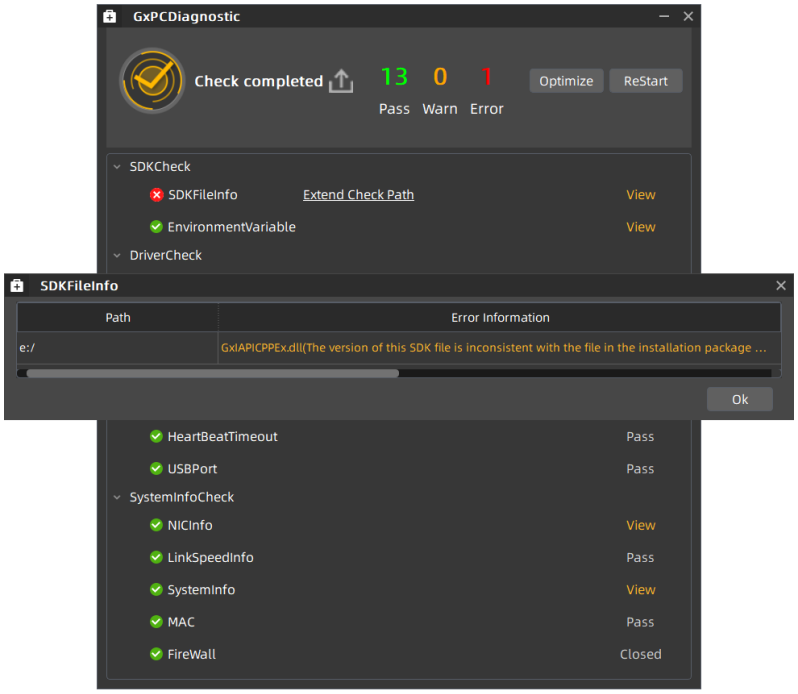



Figure 3-31 SDK Version Check Failed

**EnvironmentVariable:** This check verifies whether the Galaxy installer correctly registered environment variables, ensures Galaxy-related paths exist in both GenTL and PATH variables, and confirms non-empty files/folders in these paths. The check fails if any condition is unmet. Click the "ViewAndFix" button next to "EnvironmentVariable" to diagnose and fix errors (Figure 3-32).

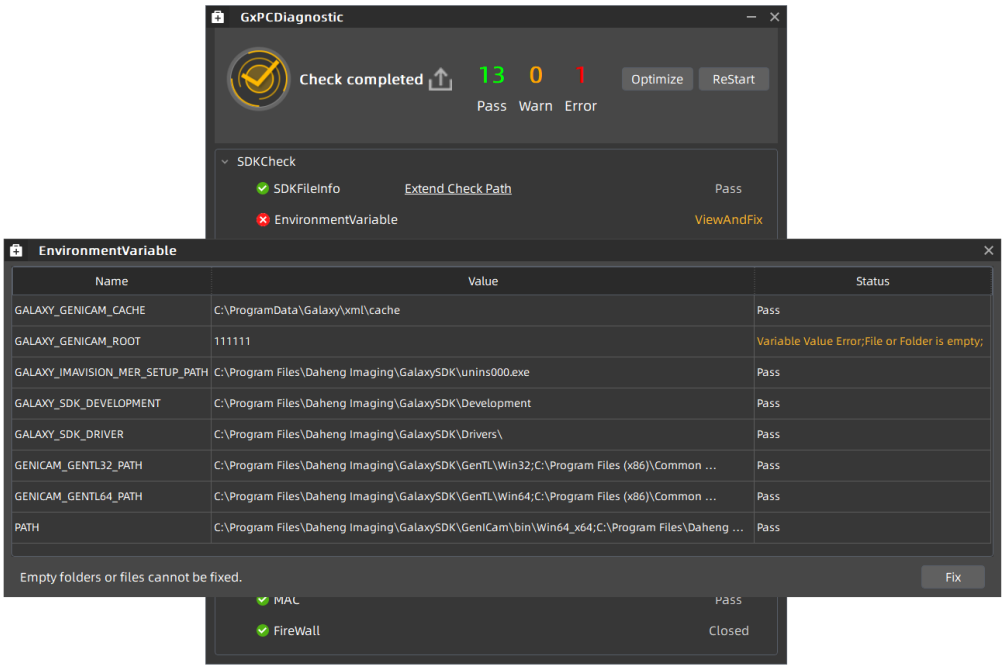



Figure 3-32 EnvironmentVariable Check Failed

**GEVDriver:** This check verifies the installation status of the network filter driver. If uninstalled, it prompts you to install the driver. Click the "Install" button next to "GEVDriver" to proceed, as shown in Figure 3-33.

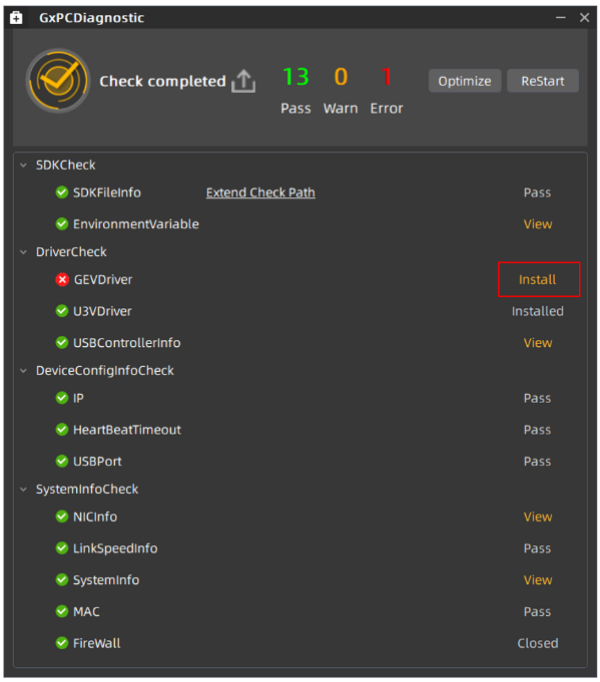



Figure 3-33 GEV Driver Check Failed

**U3VDriver:** This check verifies the installation status of USB and U3V drivers. If uninstalled, it prompts you to install them. Click the "Install" button next to "U3VDriver" to proceed, as shown in Figure 3-34.

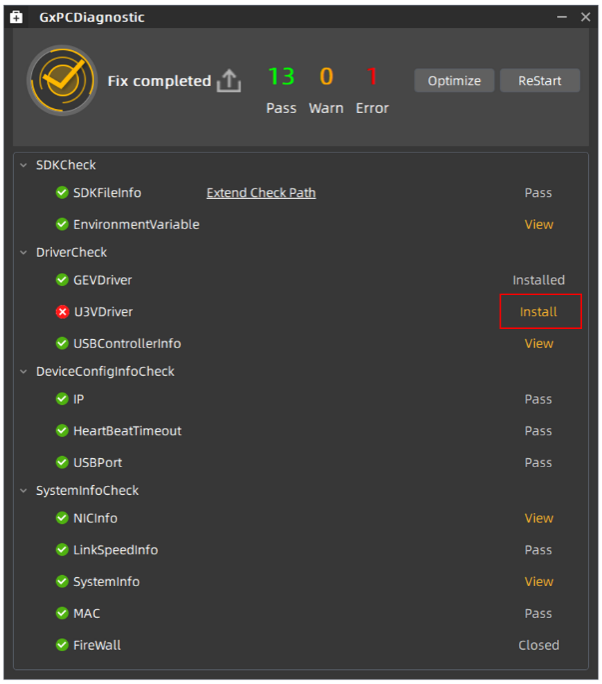



Figure 3-34 U3V Driver Check Failed

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**USBControllerInfo:** This check displays driver details for USB root controllers, including DriverDesc, ProviderName, DriverDate, and DriverVersion. Click the "View" button next to "USBControllerInfo" to see full specifications (Figure 3-35).

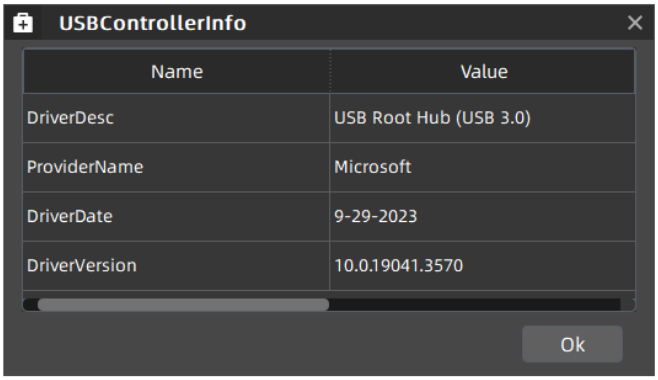



Figure 3-35 USBControllerInfo

**IP:** This check verifies whether detected camera IP addresses are on the same subnet as the host NIC. If any camera IP resides outside the subnet, the check fails. Click the "View" button next to "IP" to diagnose mismatches (Figure 3-36).

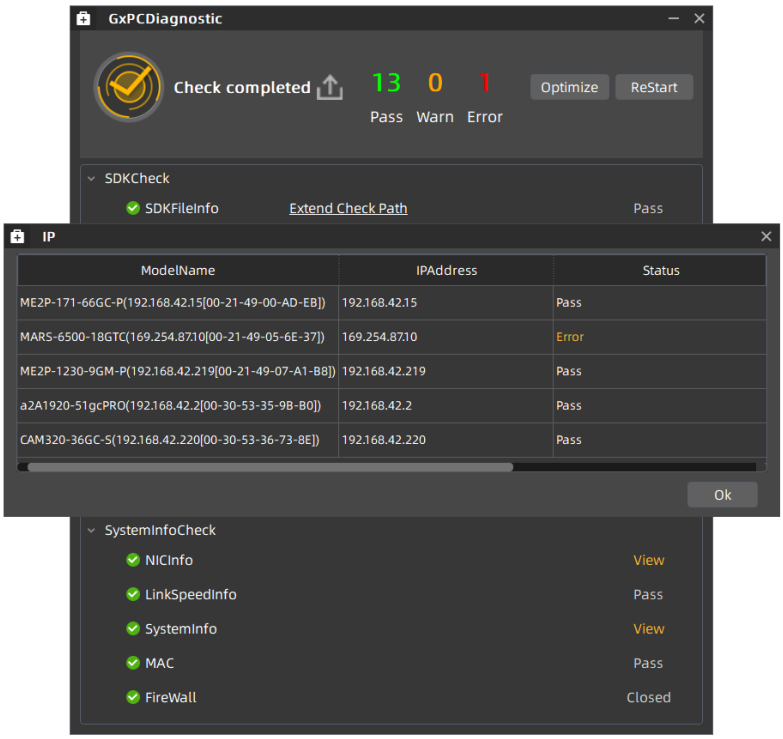



Figure 3-36 IP Check Failed

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**HeartBeatTimeout:** This check verifies if all detected network cameras have a heartbeat timeout value ≥3s. The check fails if any camera falls below this threshold. Click the "View" button next to "HeartBeatTimeout" for details (Figure 3-37). Reducing timeout below 3s is not recommended under high CPU load.

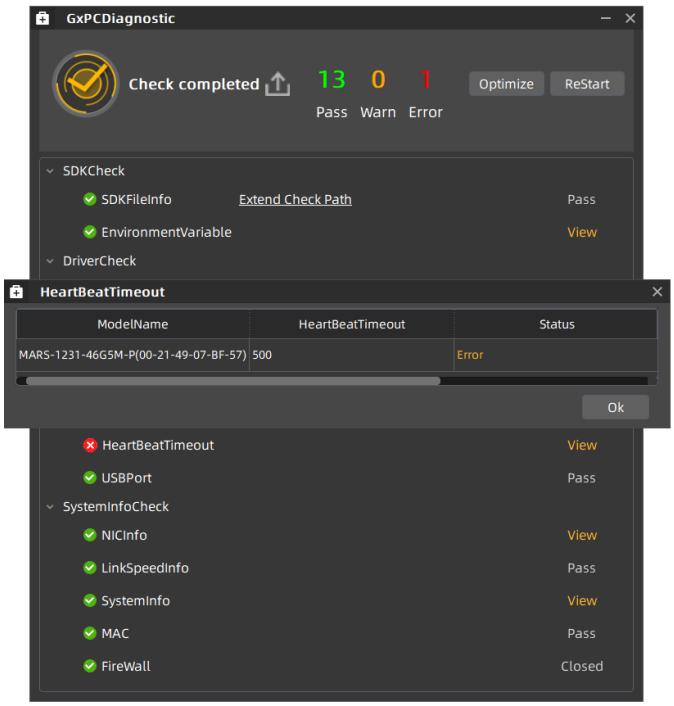



Figure 3-37 HeartBeatTimeout Check Failed

**USBPort:** This check verifies if all USB cameras are connected to correct ports (e.g., U3D cameras require USB3 ports). Detection fails for any mismatch. Click the "View" button next to "USBPort" for details (Figure 3-38).

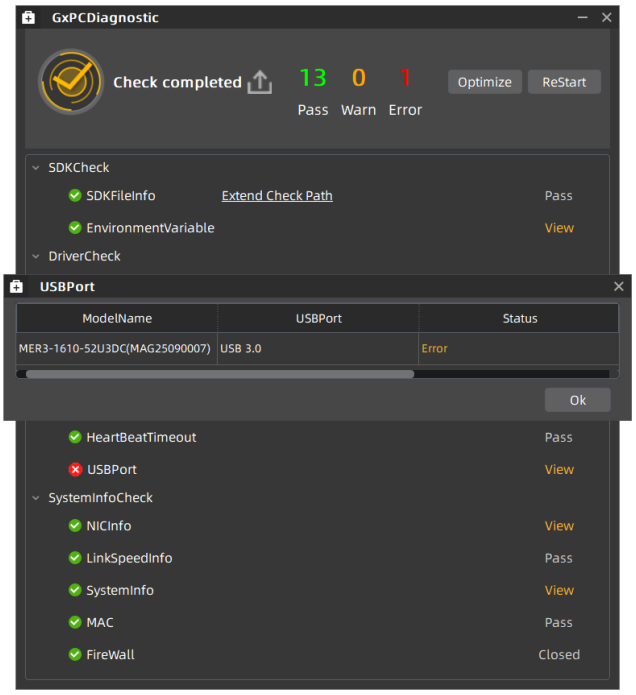



Figure 3-38 USBPort Check Failed

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**NICInfo:** This check verifies network adapter configurations (e.g., jumbo frame enablement). Fails if any setting is incorrect. Click the "ViewAndFix" button next to "NICInfo" to diagnose and fix errors (Figure 3-39).

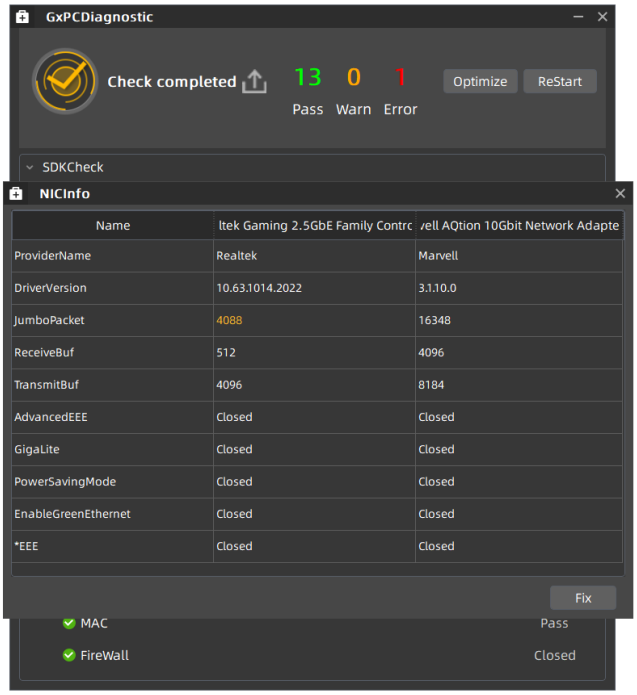



Figure 3-39 NICInfo Check Failed

**LinkSpeedInfo:** This check verifies if the network adapter's link speed matches the connected camera's required speed (e.g., 10GigE cameras require 10Gbps+ ports). Fails on any mismatch. Click the "View" button next to "LinkSpeedInfo" for details (Figure 3-40).

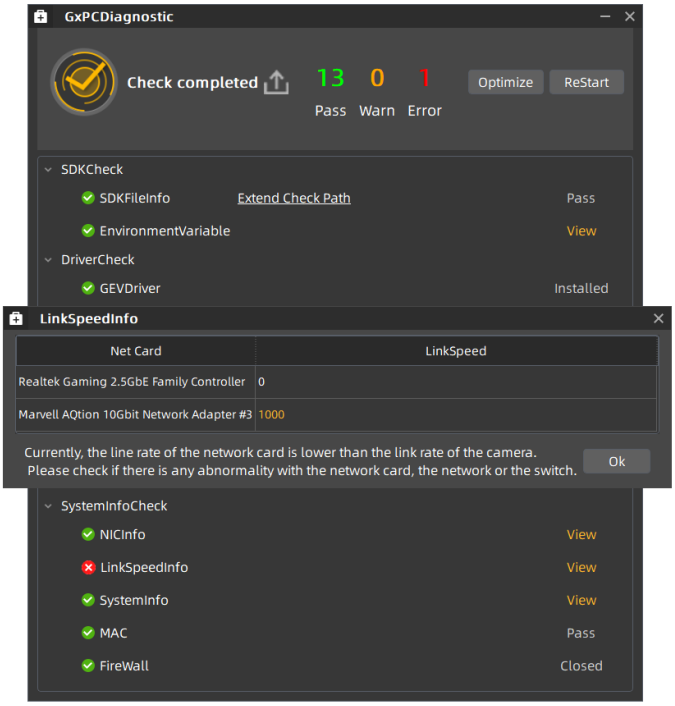



Figure 3-40 LinkSpeedInfo Check Failed

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**SystemInfo:** This check retrieves host details—including Processor, Processor cores, Available Memory, Byte Order, Operating System, Graphics Chip, and Screen Resolution—to facilitate troubleshooting. Click the "View" button next to "SystemInfo" to see full specs (Figure 3-41).

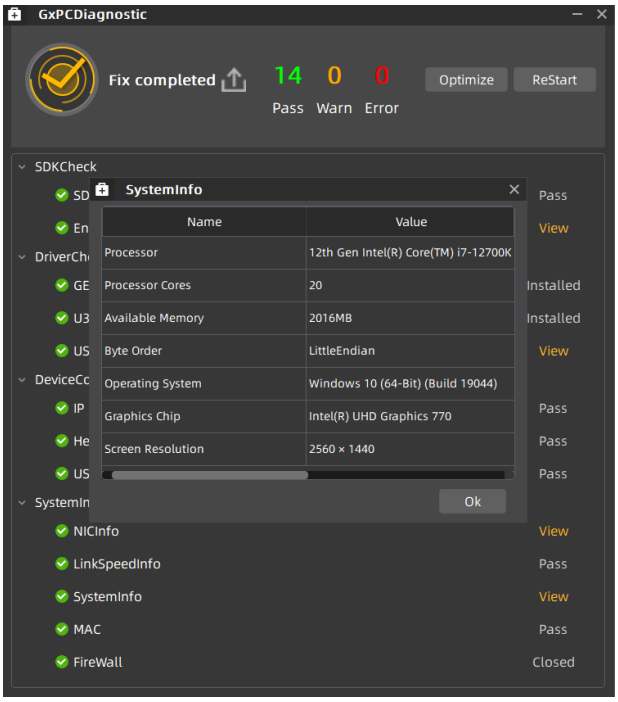



Figure 3-41 SystemInfo

**MAC:** This check detects conflicts among host NIC MAC addresses. Fails if conflicts exist. Click the "View" button next to "MAC" to see details (Figure 3-42).

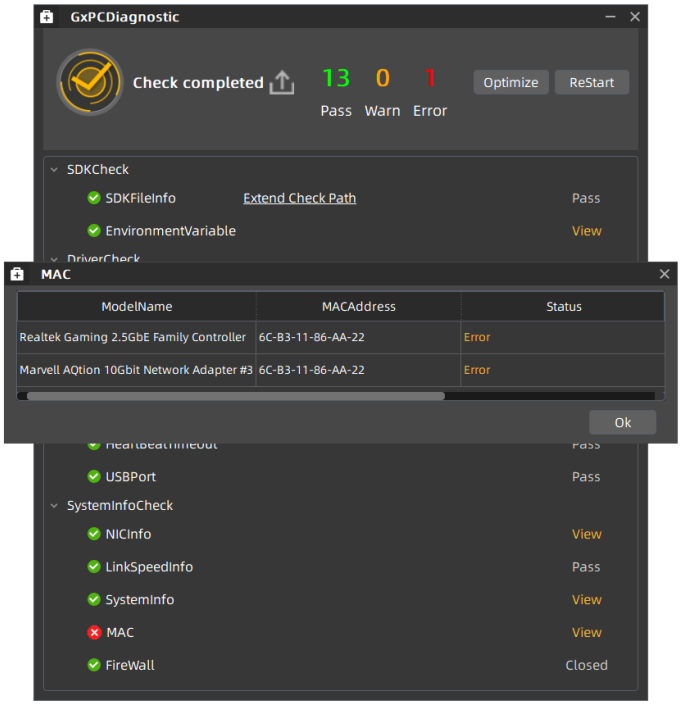



Figure 3-42 MAC Check Failed

**Firewall:** This check verifies whether the host firewall is enabled. If enabled, a warning is displayed. While active, the tool cannot perform repairs—the user must manually disable it. See Figure 3-43.

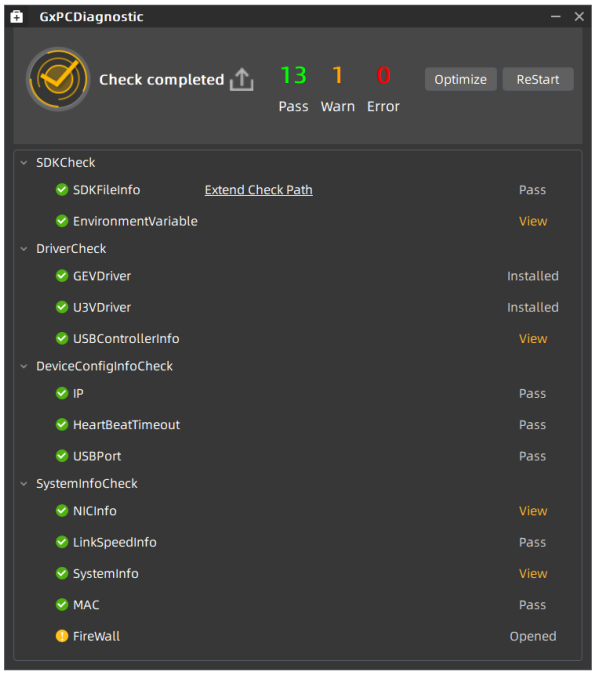



Figure 3-43 FireWall Warning

**Warning Items:** Post-scan, yellow icons indicate warnings. These may not immediately disrupt camera access but must be investigated if connection issues arise. The tool cannot auto-resolve warnings.

**Critical Items**: Red icons indicate critical failures. Some (e.g., EnvironmentVariables) can be fixed by the tool; others (e.g., USBPort) require manual fixes. Click "View" for error details. Repair methods:

1) Single-Item Repair: Click "ViewAndFix" on the item to diagnose and fix. (see Figure 3-44).

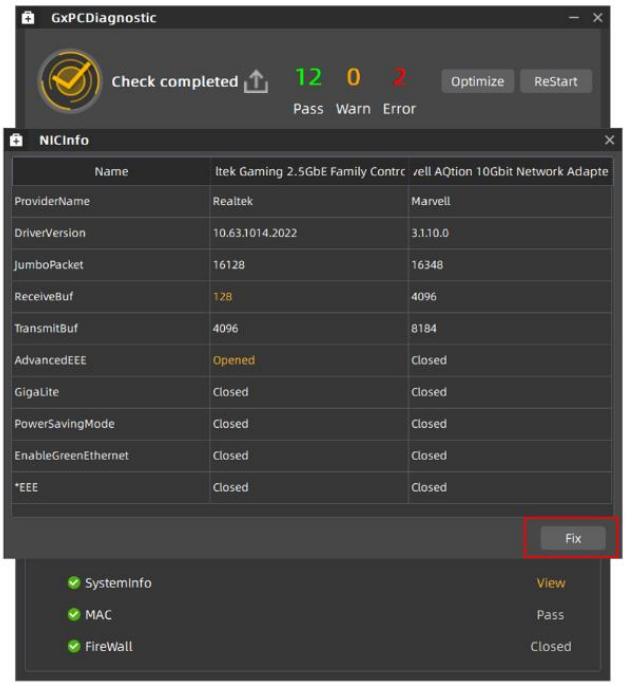



Figure 3-44 Single-Item Repair

2) Batch Repair: Click "Optimize" on the main UI to fix all repairable items (see Figure 3-45).

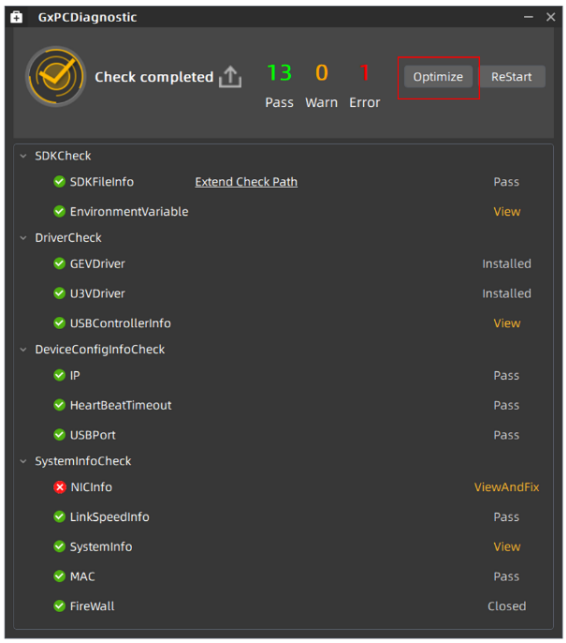



Figure 3-45 Batch Repair

**Export Logs:** After diagnosis, click the export button (top-left) to save a full report —including inspection results, SDK logs and OS event logs—as shown in the Figure 3-46.

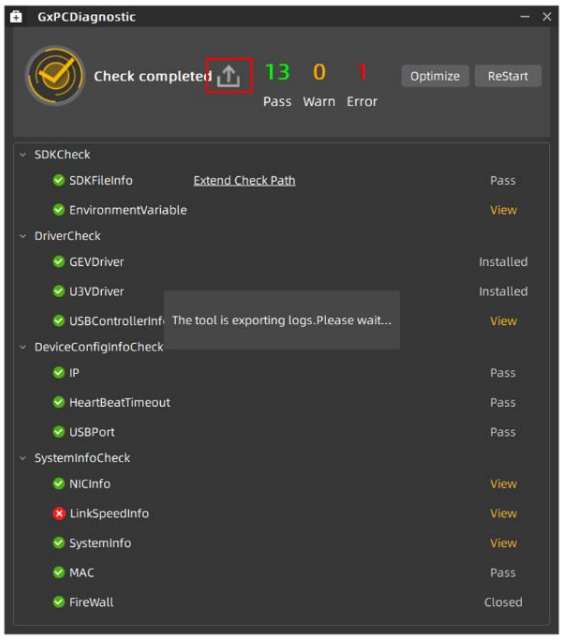



Figure 3-46 Export Logs

## 3.6. Help

**Language:** Simplified Chinese or English (Effective after restart).

**Development portal:** Quickly access SDK development path.

**Open user instructions:** Open use manual.

**About:** Check the current application and SDK version, SDK copyright information.

# 4. Tool Bar

## 4.1. Preview

1. Click “  ”.
2. Choose a file (bmp, jpg, png, tiff, raw)

**Instruction:**

1. The following formats can be previewed directly: bmp, jpg, png, tiff.
2. Raw image can be opened only in right format: “ xxx_width_height_format.raw ” , or there will be an error prompt until you correct it.

## 4.2. Full Screen

Click “  ” to full screen the window (Still remain tile even multiple windows after clicking).

## 4.3. Acquire Information

Click “  ” to open the Acquire Information Status window, see as Figure 4-1:

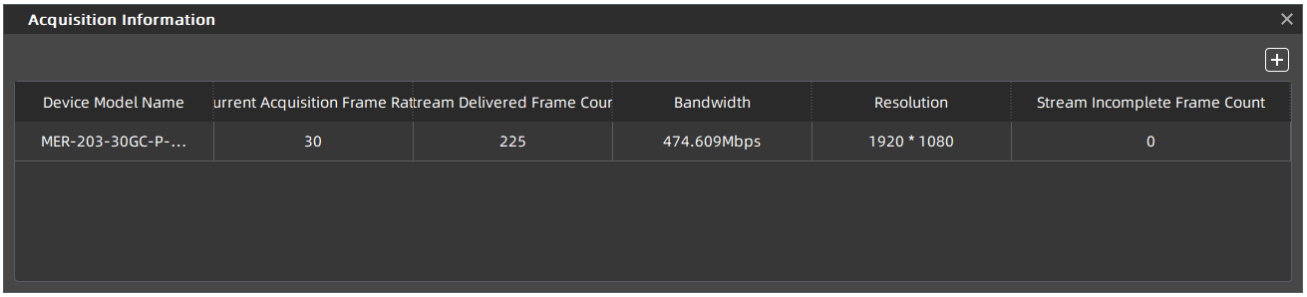



Figure 4-1 Acquire Information Status

Users can add more items by clicking “ + ” in the top right-hand corner, see as Figure 4-2

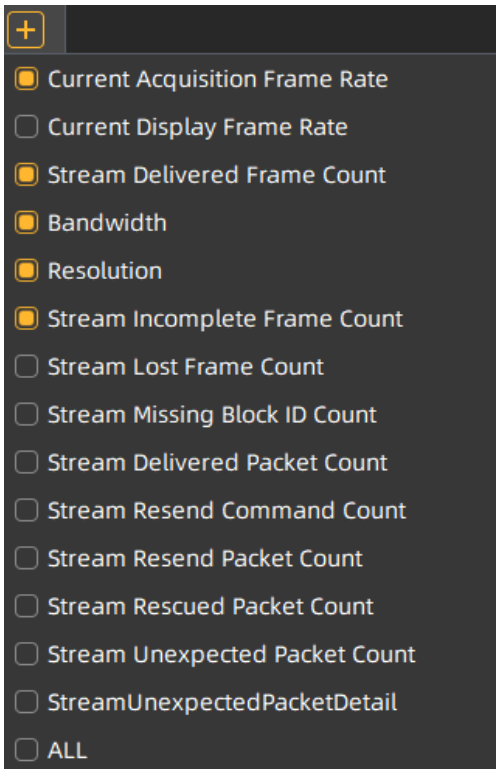



Figure 4-2 Information options

Note: The number of received frames here refers to the number of frames received by the camera, while the frame count under the camera window refers to the number of frames displayed on the interface.

## 4.4. Show All Preview

 : The display interface of the opened camera will switch to the display state.

## 4.5. Hide All Preview

 : The display interface of the opened camera will switch to no display state.

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# 5. Device List

## 5.1. Shortcut Functions

**Open all devices:** Click “  ” to connect all searched cameras in batch.

**Close all devices:** Click “  ” to disconnect all connected cameras in batch.

**Start acquisition all devices:** Click “  ” to start all cameras acquisition in batch.

**Stop acquisition all devices:** Click “  ” to stop all cameras acquisition in batch.

**Open device:** Click “  ” to open the selected camera, which equals to double click the camera.

**Close device:** Click “  ” to disconnect the selected camera.

**Start acquisition:** Click “  ” to start acquire the current camera.

**Stop acquisition:** Click “  ” to stop acquire the current camera.

## 5.2. Device Status

Device List is divided into GigE, USB2.0, USB3.0 and CXP four types according to the device interface. All searched cameras will be listed in the Device List area, the different icons represent the different status of different interface cameras.

GigE camera status description, see as Table 5-1:

| Icons | Status | Descriptions |
|---|---|---|
|  | Available | Double click or through **Open device** to connect the camera |
|  | Connected | The camera is connected and operable |
|  | Acquiring | The camera is connected and in acquisition status |
|  | Occupied | The camera is connected by other device so failure in connecting with the current client, please disconnect other device to connect the current client |
|  | Unreached | The camera is unreachable in the LAN, please modify the IP to the same network segment to use it |
|  | Read Only | When the camera is opened in Control mode by other processes. |
|  | Unknown | Cannot get the current accessible state for some reason |

Table 5-1 GigE camera status

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USB3.0 camera status description, see as Table 5-2:

| Icons | Status | Descriptions |
|---|---|---|
|  | Available | Double click or through **Open device** to connect the camera |
|  | Connected | The camera is connected and operable |
|  | Acquiring | The camera is connected and in acquisition status |
|  | Occupied | The camera is connected by other device so failure in connecting with the current client, please disconnect other device to connect the current client |

Table 5-2 USB3.0 camera status

USB2.0 camera status description, see as Table 5-3:

| Icons | Status | Descriptions |
|---|---|---|
|  | Available | Double click or through **Open device** to connect the camera |
|  | Connected | The camera is connected and operable |
|  | Acquiring | The camera is connected and in acquisition status |
|  | Occupied | The camera is connected by other device so failure in connecting with the current client, please disconnect other device to connect the current client |

Table 5-3 USB2.0 camera status

CXP camera status description, see as Table 5-4:

| Icons | Status | Descriptions |
|---|---|---|
|  | Available | Double click or through **Open device** to connect the camera |
|  | Connected | The camera is connected and operable |
|  | Acquiring | The camera is connected and in acquisition status |
|  | Occupied | The camera is connected by other device so failure in connecting with the current client, please disconnect other device to connect the current client |

Table 5-4 CXP camera status

## 5.3. More Functions

**NIC Settings:** Click a NIC, and open the NICConfigTool to set the properties, see details in 3.5.4NICConfigTool.

**Add Remote Device:** Input the remote device IP address to add new device.

**Auto Update:** Update the Device List automatically in every 5s or click update button to update device manually.

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# 6. Preview

## 6.1. Shortcut Functions

**Start/Stop Acquisition:** Click “  ” to start acquisition, click “  ” to stop acquisition.

**Start/Stop Preview:** Click “  ” to hide the image, click again to show it.

**Save the current image:** Click “  ” to save the current image, if you want to save the image settings as well, see details in 3.4.2 Save Image Setting.

**Show Crosshair:** Click “  ” to use the crosshair and click again to close it, hover over the button to edit panel, see Figure 6-1.

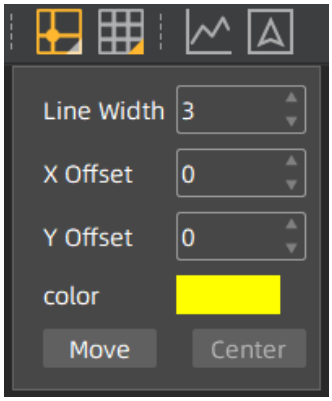



Figure 6-1 Crosshair edit panel

**Show Grid:** Click “  ” to open the grid and click again to close it, hover over the button to edit panel, see Figure 6-2.

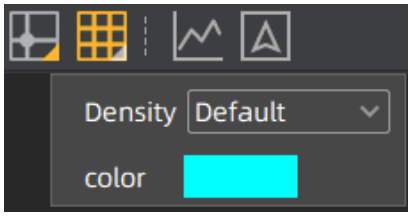



Figure 6-2 Grid edit panel

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**Histogram:** Click “  ” to the Histogram Window, and click again to close it.

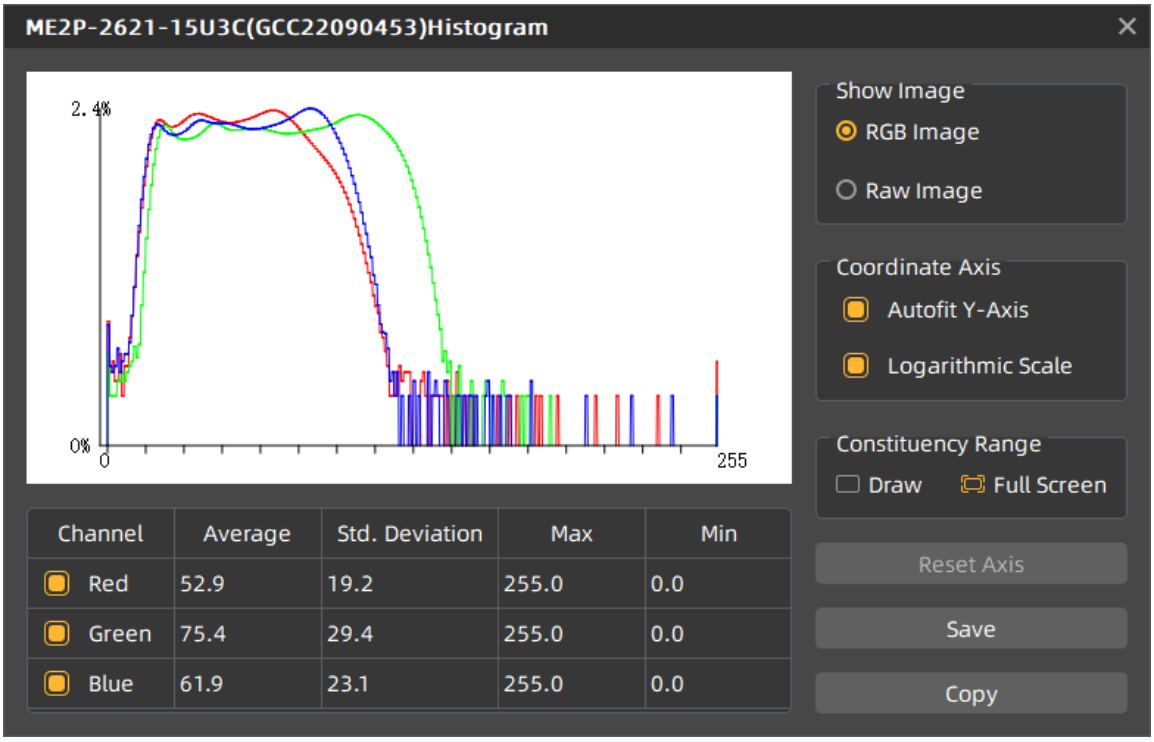



Figure 6-3 Histogram of color camera

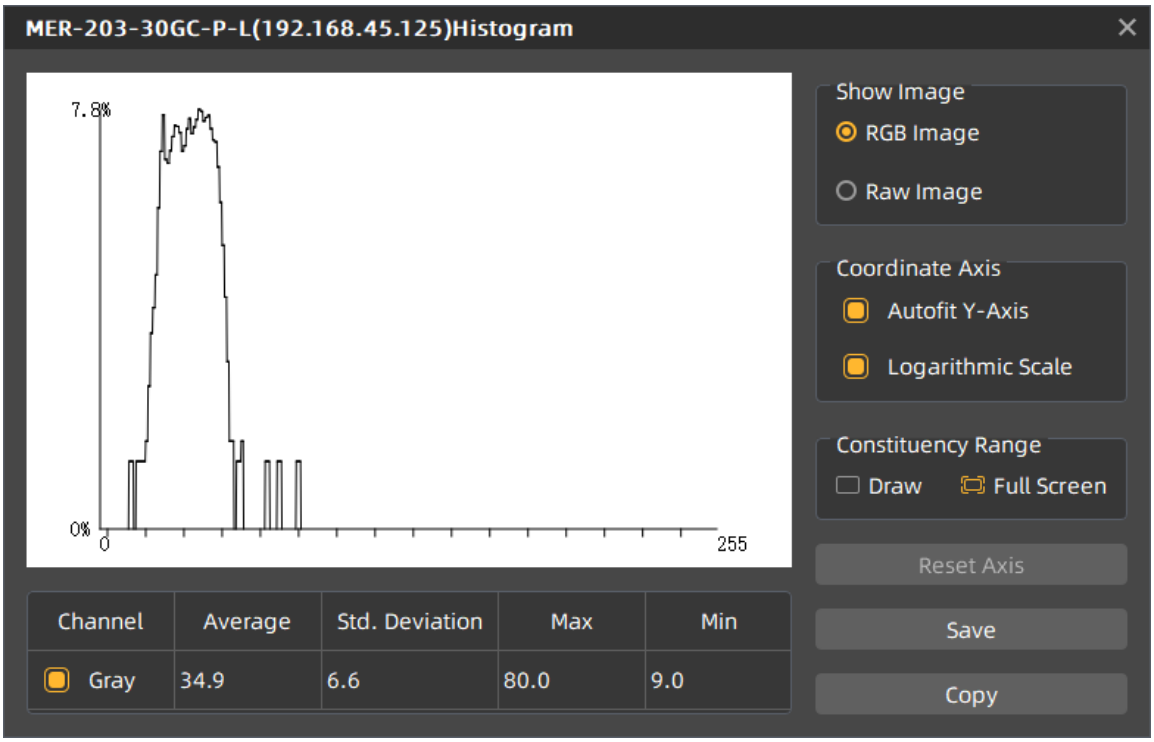



Figure 6-4 Histogram of mono camera

1. Choose Channel

   Select the R/G/B channel beneath the coordinate according to your demand. Color camera have R, G, B three channels while mono camera only have one Luminance channel.

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2. Show Image

   RGB Image: Image after Interpolation process.

   Raw Image: The original image.

3. Coordinate Axis

   Here are Autofit Y-Axis and Logarithmic Scale two options can be choose.

4. Constituency Range

   Click **Draw** to set ROI (Region of Interest), which will only show the ROI histogram, click **Full Screen** to cancel ROI.

5. Axis zoom and reset

   **Ctrl** + Mouse wheel up and down in axis area to zoom the axis and click **Reset Axis** to restore the axis.

6. Save

   Save the histogram data as .csv to the local file, and click **Copy** button to copy the histogram data to clipboard.

**Sharpness:** Click "  " to open the Sharpness Window and click again to close it. This window can display the clarity of the current image in real time to achieve the auxiliary focusing function.

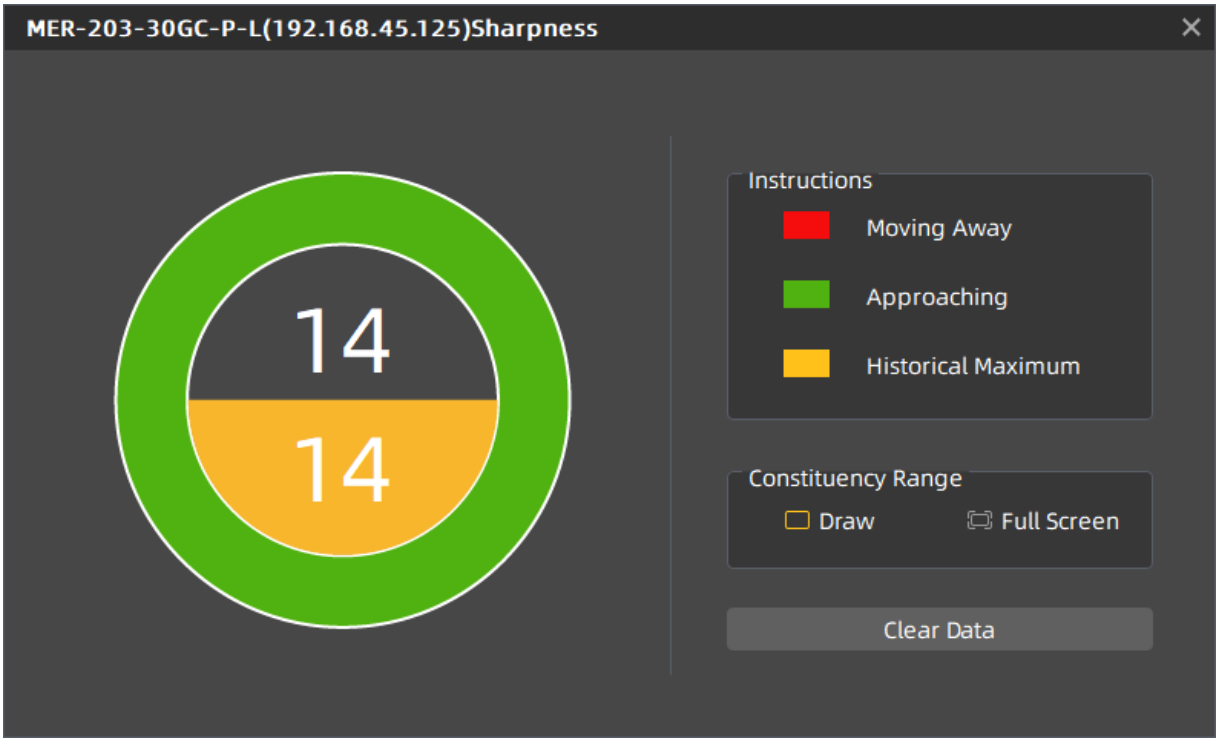



Figure 6-5 Sharpness window

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1. Sharpness Value

   The left side of the window shows the sharpness value, the yellow area means the historical maximum value and the white area means the current sharpness value. The green out-loop means the rising sharpness value and it will turns in red while falling.

2. Constituency Range

   Click **Draw** to set ROI (Region of Interest), which will only show the ROI sharpness value, click **Full Screen** to cancel ROI.

3. Clear Data

   Click **Clear Data** to zero the historical maximum sharpness value.

**RGB to Mono:** Click "  " to change the image from color to mono, click again to cancel. Mono camera without this function.

**Acquisition Status Info:** At the bottom of the preview window is the real-time information of the acquisition image, which can be selected for display through the configuration button in the lower right corner.

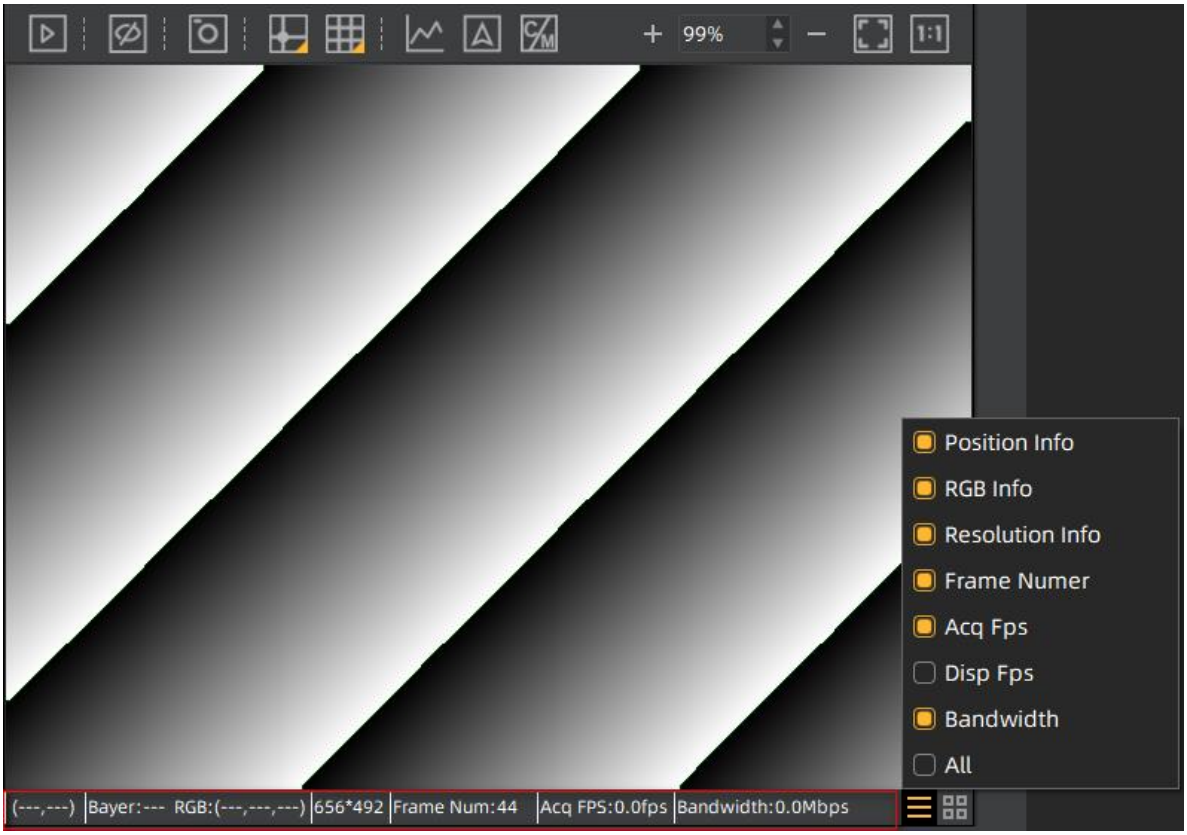



Figure 6-6

**Display Settings:** Click the button in the bottom right-hand corner of the Preview Window, see as follow:

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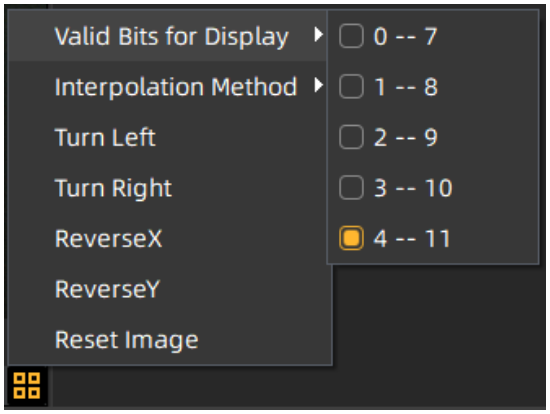



Figure 6-7 Display Settings

Valid Bits for Display: For Mono8, there is only 0 – 7 can be selected. For Mono10, there are 0 – 7, 1 – 8 and 2 – 9 (by default) three options can be selected. For Mono12, 4 – 11 is by default.

Interpolation Method: There are 2×2 Neighbour(Faster), 3×3 Neighbour(Moderate) and 5×5 Adaptive(Better) three options.

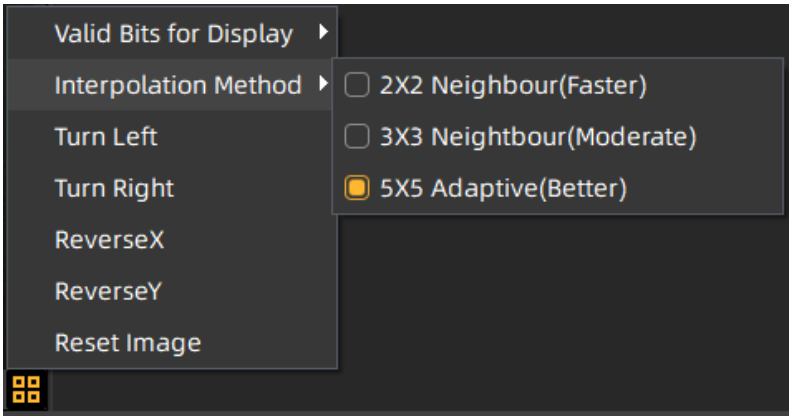



Figure 6-8Interpolation Method

Zoom In and Out: Support the size scaling function of images. You can control the zoom ratio by pressing **Ctrl + mouse scroll** in the preview window, or right-click in the preview window or configure it in the upper right corner of the preview window.

Rotate Mirror: Support the setting of image rotation 90 degrees left, 90 degrees right, horizontal mirroring, and vertical mirroring functions, also can rotate or mirror the image with one click to reset. The interface of Figure 6-9 can be shown by right-clicking the preview window or clicking the **Settings** button in the lower right corner of the preview window.

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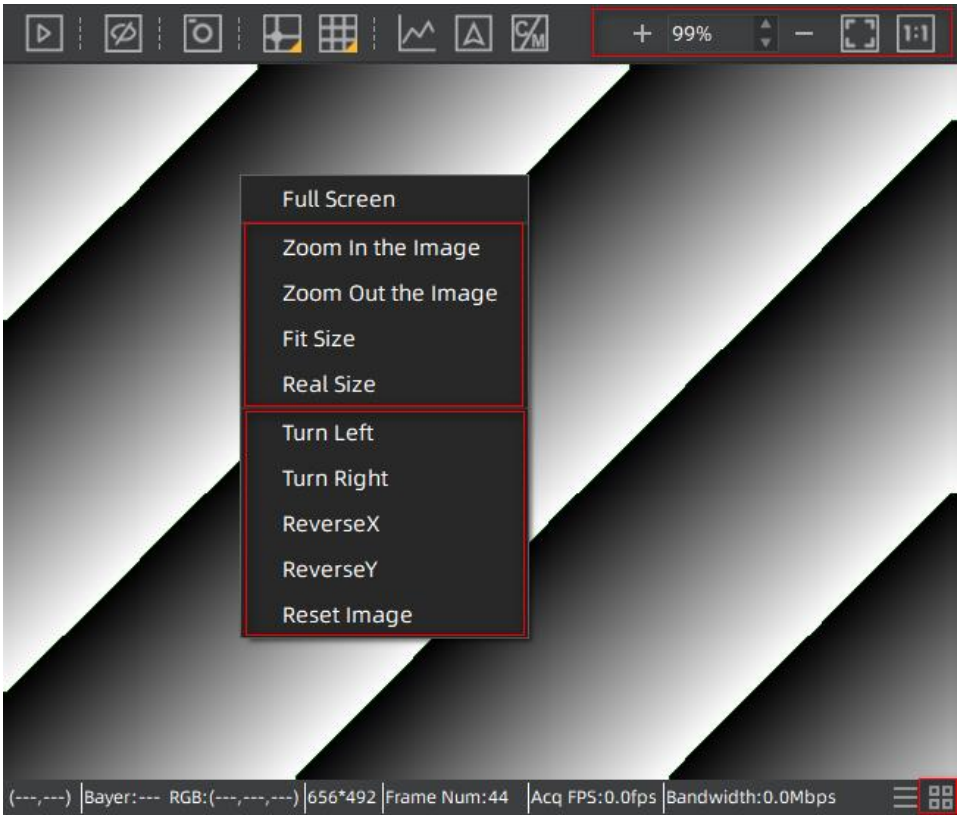



Figure 6-9

## 6.2. Preview

- Steps

1. Connect the camera in the Device List.
2. Click the shortcut button “ ” above the Device List or “ ” button of the Preview Window to start acquisition.
3. Click “ ” to hide the preview window if you do not have to preview the image in real time.
4. Preview Window will auto arrange the layout if there is a need to open multiple cameras to preview the image.

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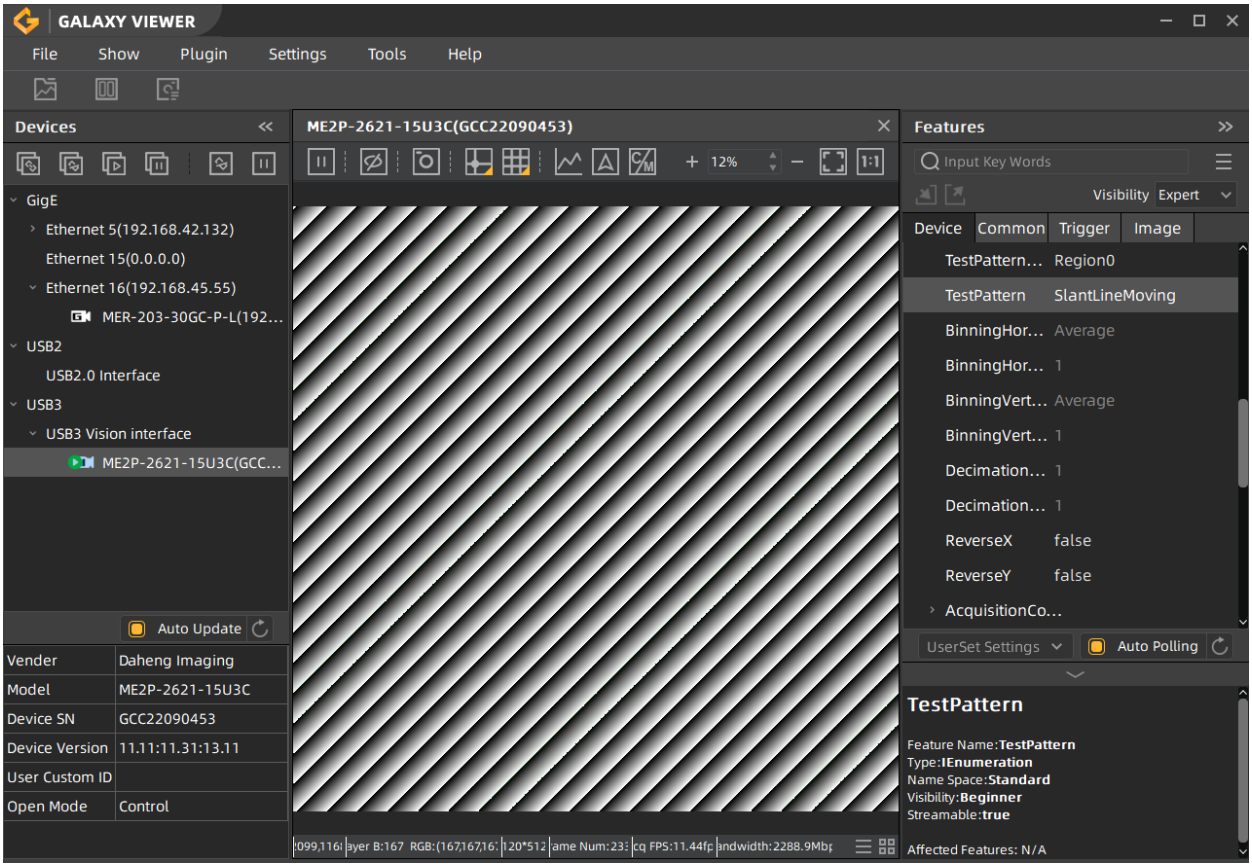



Figure 6-10 Single image preview

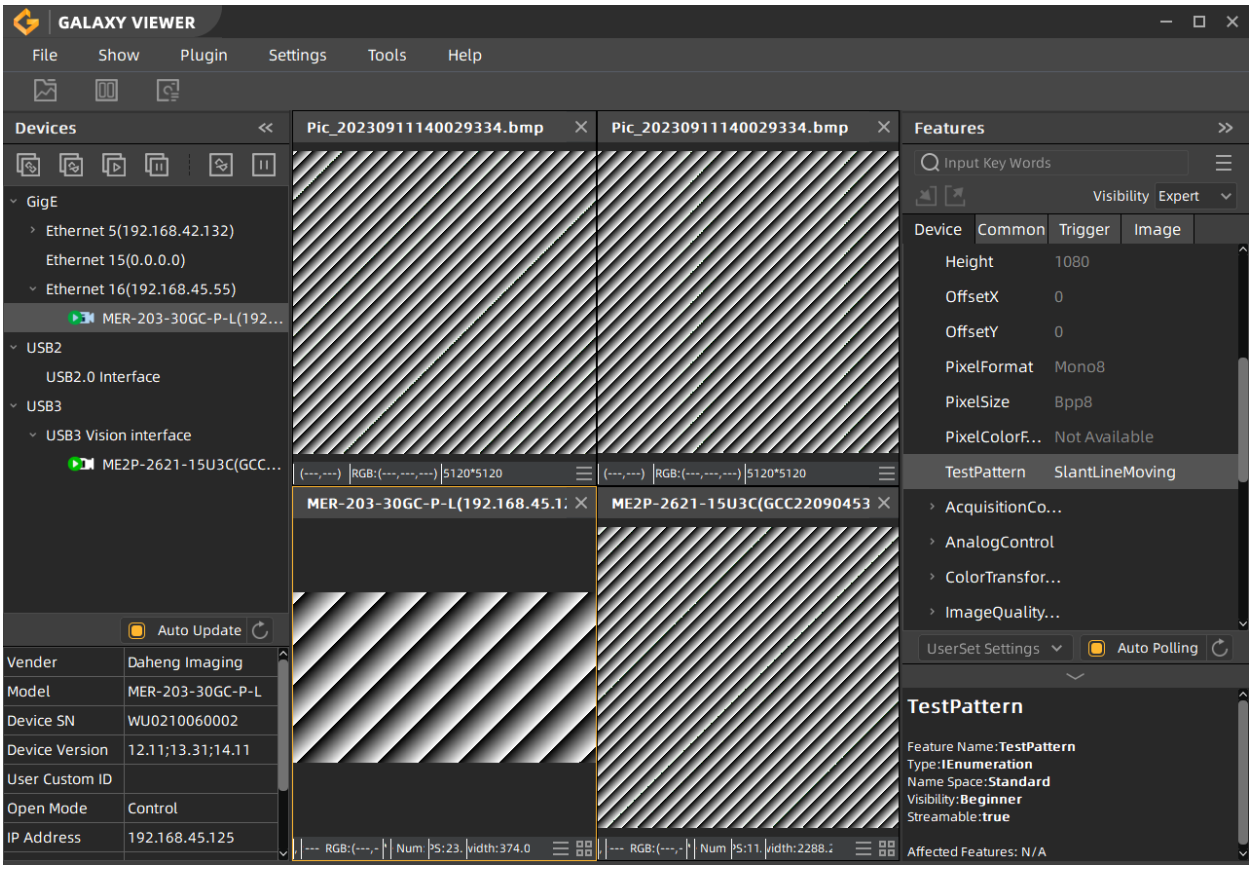



Figure 6-11 Multiple images preview

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# 7. Features

Including **Device**, **Common**, **Trigger** and **Image** settings.

## 7.1. Common

Including the specific attributes of the connected camera.

The Common List is hid while disconnecting and viewable while connecting.

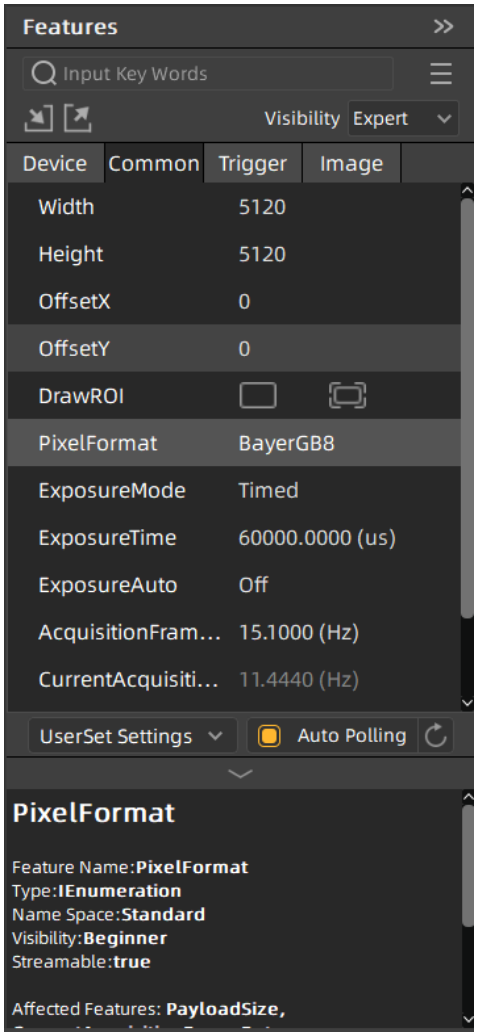



Figure 7-1 Common List

**Input Key Words:** Input the keywords you want to search and select in the drop-down box, next it will automatically jump to the corresponding item.

**Favorites:** Users can put the common items into Favorites.

**Export Device Settings:** Click “ ” to export the camera configuration files to the local file.

**Import Device Settings:** Click “ ” to import the file into camera.

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**Visibility:** Here you can choose Beginner, Expert or Guru mode, and the corresponding attributes will show as below.

**UserSet Settings:** Load User Set or Save as Userset0 and other functions.

**Feature Information:** Show the corresponding attributes details below. This area is collapsible.

## 7.2. Quick Property Page

Which is used for users to look up the camera attributes quickly by classifying the common attributes, including Common, Trigger and Image.

**Common:** Including Width, Height, OffsetX, OffsetY, PixelFormat, ExposureMode, ExposureTime, ExposureAuto, Gain, GainAuto, AcquisitionFrameRate, CurrentAcquisitionFrameRate. Users can set ROI by clicking Draw button, and click Full Screen to the max resolution.

**Trigger:** Including TriggerMode, TriggerSource, TriggerSoftware, TriggerActivation, TriggerDelay, LineSelector, LineMode, LineSource, UserOutputSelector, and UserOutputValue.

**Image:** Including GammaEnable, GammaMode, Gamma, SharpnessMode, Sharpness, BalanceRatioSelector, BalanceRatio and BalanceWhiteAuto.

**Instruction:** If a hand-drawn ROI operation is performed while the camera is in AcquisitionStop status, then the next time you draw will still on the previous image, because of the image is not updated during AcquisitionStop, so, the drawing area may exceed the ROI setting range, which will result in ROI setting failure. If you want to accurately execute the hand-drawn ROI, it is recommended to operate in the AcquisitionStart status. Alternatively, update the image after setting the ROI, and then draw the ROI.

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# 8. Feature Document

Feature Document is at the bottom of the interface, and shows the API information of C, C++, and C# users. See Figure 8-1

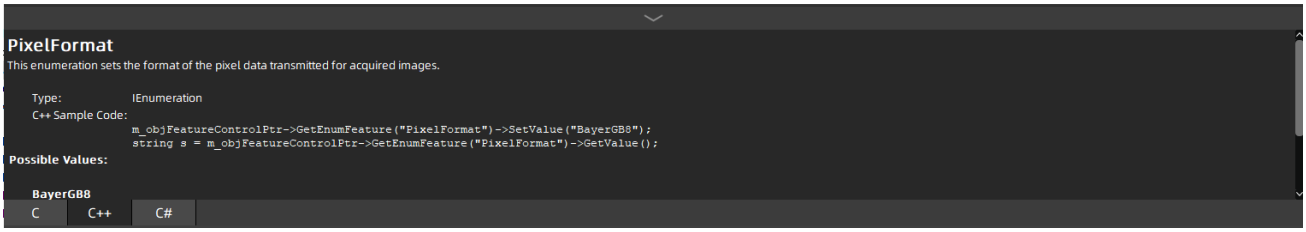



Figure 8-1 Feature Document

# 9. FAQ

Please look up the following table to solve your problems, if still existing, send your problems to our technicians: support@daheng-imaging.com.

| Problem Description | Possible Reasons | Answers |
|---|---|---|
| Unable to search the GigE camera. | 1. The camera is not powered.<br>2. Abnormal network connection.<br>3. Camera and NIC are not in the same network seg3-2ment. | 1. Judge the power conditions by the LED colors.<br>2. Check the camera IP status by GxGigEIPConfig tool. |
| Unable to search the U3 camera. | 1. USB cable is abnormal.<br>2. No U3 drive installed. | 1. Judge the power conditions by the LED colors.<br>2. Check U3 drive installation status through GxDriverManager tool. |
| GigE camera: OpenDriver failed. | 1. There is no filter drive installed in GigE cameras. | 1. Check GigE Driver installation status through GxDriverManager tool. |
| GigE camera: device is denied. | 1. GigE camera is opened by other process.<br>2. The heartbeat time of GigE camera is 5min, still not yet reset. | 1. Check the camera current status first, and if you still want to open the camera, please double click the camera through GxGigEIPConfig tool to release it of occupation. |
| U3 camera opening error. | 1. U3 cable quality problem, for example, use U2 when U3 is expected. | 1. Replace the cable. |
| U3 camera return error includes the following string: GenICam::USB::CSystemAPI::IoCtrl | 1. Electromagnetic interference.<br>2. Compatibility of the master controller. | 1. Add anti electromagnetic interference equipment.<br>2. Replace the master controller. Intel (Z77 chipset) or Renesas (chip version: uPD720202, driver version: 3.0.23.0) are recommended.<br>It is not recommended to use AMD or Etron master controller. |
| U3 camera return error includes the following string: Bad format of U3V acknowledgment packet from device. | 1. Compatibility of the master controller. | 1. Replace the master controller. Intel (Z77 chipset) or Renesas (chip version: uPD720202, driver version: 3.0.23.0) are recommended.<br>It is not recommended to use AMD or Etron master controller. |
| Bad preview image. | 1. Aperture is too small.<br>2. Exposure time is too short. | 1. Adjust the value of aperture and exposure time to the appropriate range. |
| Start acquisition error includes: “Failed to attach buffer”. | 1. 32 bit OS does not have enough buffer for the camera acquisition. | 1. Use 64 bit OS or reduce the buffer number. |

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| Problem Description | Possible Reasons | Answers |
|---|---|---|
| There is no image while in external trigger. | 1. Camera trigger mode is not enabled or trigger source is incorrectly.<br>2. Trigger wiring error. | 1. Open camera trigger mode, select the trigger source and lay the cables correctly. |
| Serious frame loss (GigE camera). | 1. Use megabit network. | 1. Ensure the current network speed reaches gigabit network. |
| Serious incomplete frame (GigE camera). | 1. JumboFrame mode is unable. | 1. Set the network card to JumboFrame mode and the camera to the maximum packet length. |
| No image in the preview window. | 1. Camera packet size does not match the NIC packet size. | 1. Ensure that the current packet length of the camera is less than or equal to the network card supported packet length. |

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# 10. Revision History

| No. | Version | Changes | Date |
|---|---|---|---|
| 1 | V1.0.0 | 1. Initial release | 2023-09-11 |
| 2 | V1.0.1 | 1. Add CXP and JUP series product related software descriptions | 2024-03-26 |
| 3 | V1.1.0 | 1. Add GxLogViewer and GxUpgradeTool | 2024-06-12 |
| 4 | V1.1.1 | 1. Add Adaptive image buffer caching in 3.4.1 section<br>2. Add 3.5.6.Action Command Window<br>3. Add 4.4 and 4.5 sectins | 2024-12-11 |
| 5 | V1.1.2 | 1. Update section 3.3.6 Flat Field Correction Plugin | 2025-07-21 |
| 6 | V1.1.3 | 1. Add 3.5.7 GxPCDiagnostic | 2025-12-12 |

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# 11. Contact Us

## 11.1. Contact Sales

If you need to order products or inquire product information, please contact our sales:

Tel: +86 10 8282 8878-8081

Email: isupport@daheng-imaging.com

## 11.2. Contact Support

If you have any questions in using DAHENG IMAGING products, please contact the experts on our support team:

Tel: +86 10 8282 8878

Email: isupport@daheng-imaging.com

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